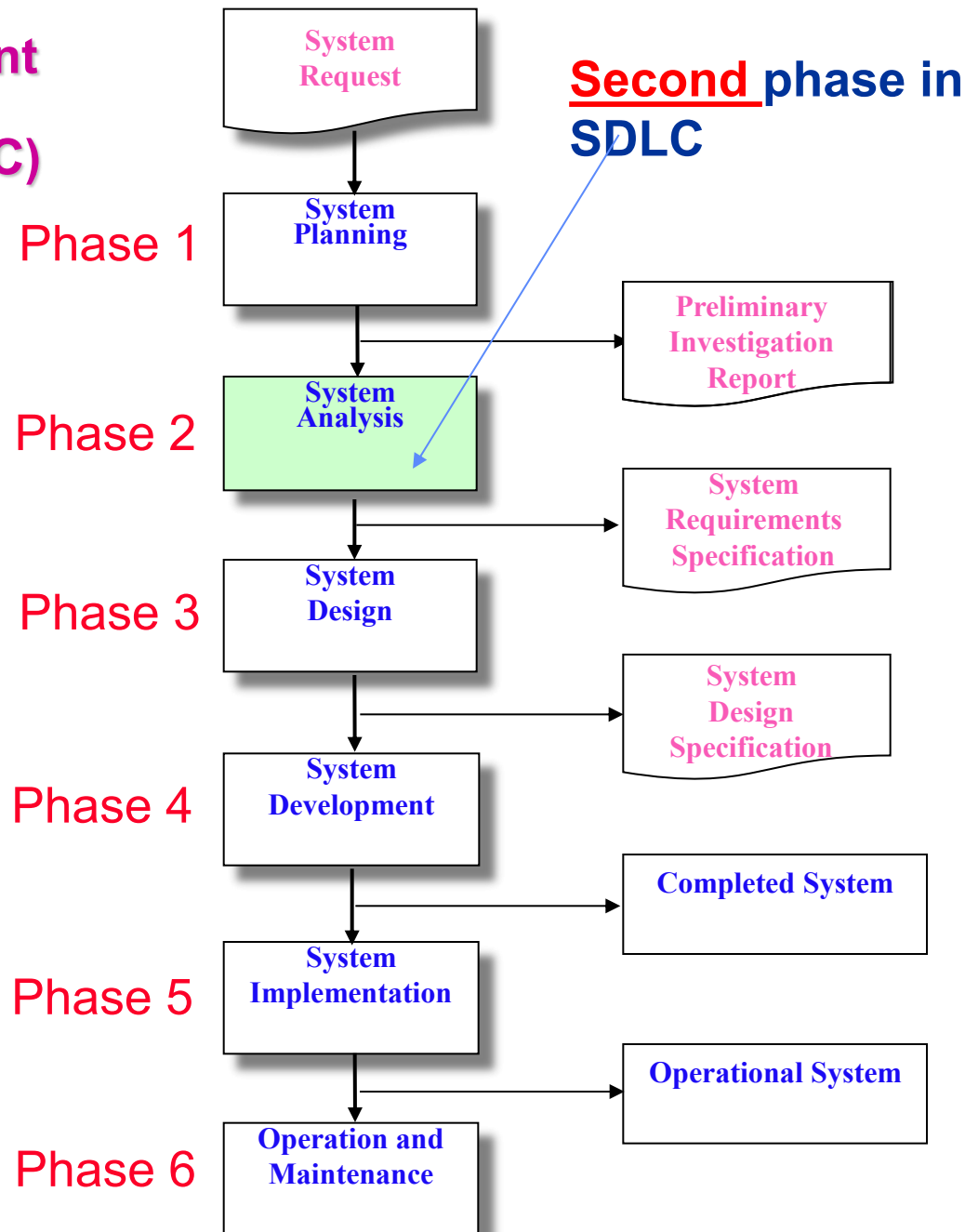


Chapter 4

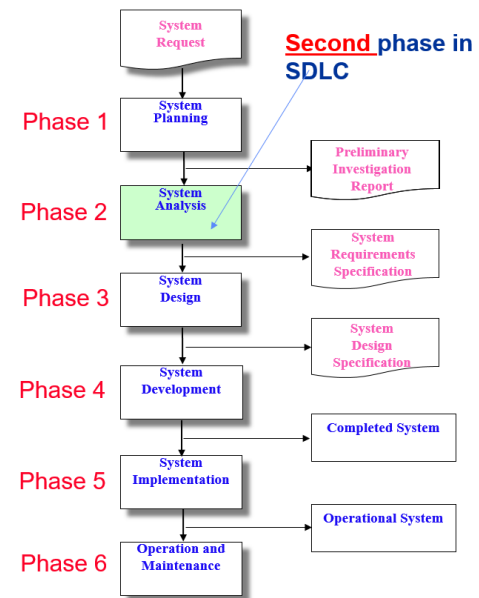
System Analysis 2 **- Fact Recording 1**

System Development Life Cycle (SDLC)



Part 2 : Fact Recording

- **Purpose** - Facts about the existing system must be recorded / documented so that they can be *referred* to during the subsequent stages.
- **Methods** - System analyst may use *different* recording *methods* eg descriptions, diagrams etc



Models Used in Fact Recording

1. Narrative Description (most popular)
2. Data Flow Diagram (**DFD**)
3. Entity Relationship Diagram (**ERD**)
4. Functional Decomposition Diagram (**Structure Chart**)
5. **Decision Tables**
6. **Decision Tree**
7. Flowcharts eg. System flowchart, program flowchart, clerical procedural flowchart.
8. Others eg UML diagrams (eg Activity Diagrams)

Examples of Models Used in Analysis Phase

The analysis phase activity may involve creating a variety of **models**.

They are referred to as **logical** models. Describe *what* is required without committing to one specific *technology* on *how* to carry out.

Consider an order processing system. Whenever an order is received the system should first check to determine if payment has been received with the order. There are two possibilities - yes and no - which lead on to differing responses.

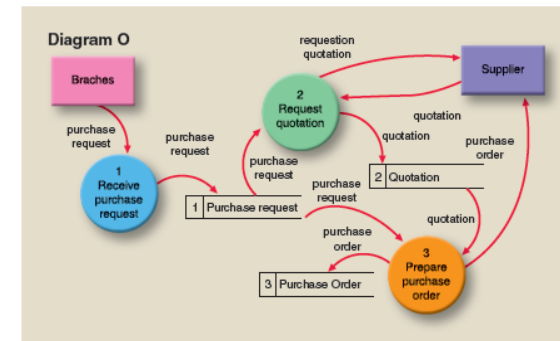
If payment is included with the order, the system would then have to check on the availability of the items ordered. Here there are two further possibilities - the items can be in stock or not.

If the item are in stock they would then be shipped or delivered to the customer.

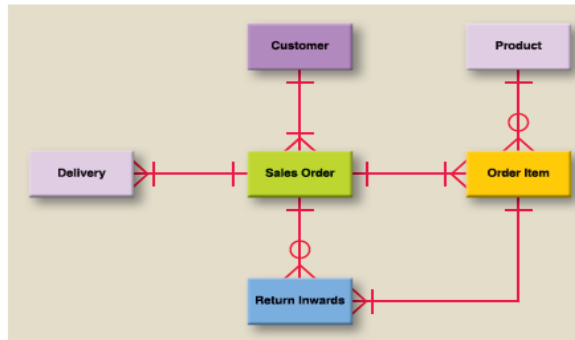
If the item is not in stock a record of the customer's order orders to be made so that the item will be despatched when stock becomes available.

If the customer has not included payment with order a similar set of activities would be undertaken but they would be preceded by credit evaluations.

Narrative Description



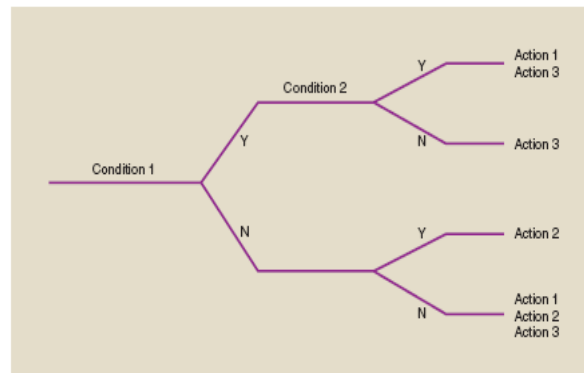
Data Flow Diagram



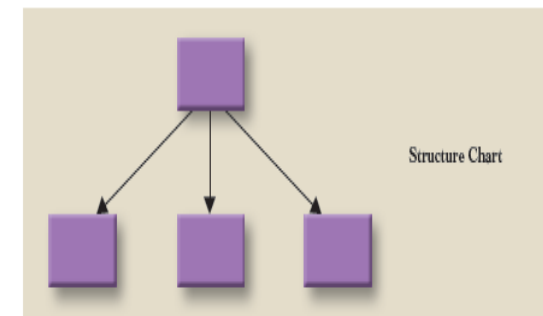
Entity Relationship Diagram

Conditions	1	2	3	4	5	6
Is the order < \$10?	Y	N	N	N	N	N
Is the credit rating good?		Y	N	Y	Y	Y
Is the order \$10 - \$100?		Y		N	Y	N
Customer > 5 years?		N		N	Y	Y
Approve order	X	X		X	X	X
Give 0% discount	X					
Give 3% discount		X			X	
Give 5% discount				X		X
Add 1% to discount					X	X
Refer to supervisor			X			

Decision Table



Decision Tree



Structure Chart

1. Narrative Description

.... in words

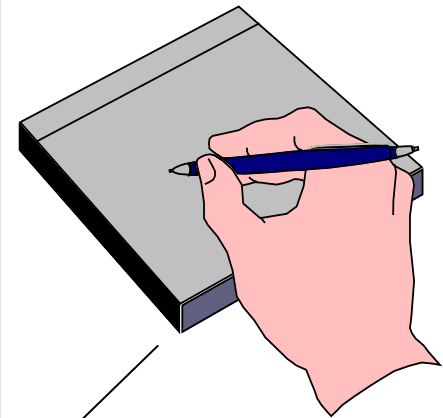
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If payment is included with the order, the system would then have to check on the availability of the items ordered. Here there are two further possibilities – the items can be in stock or not.

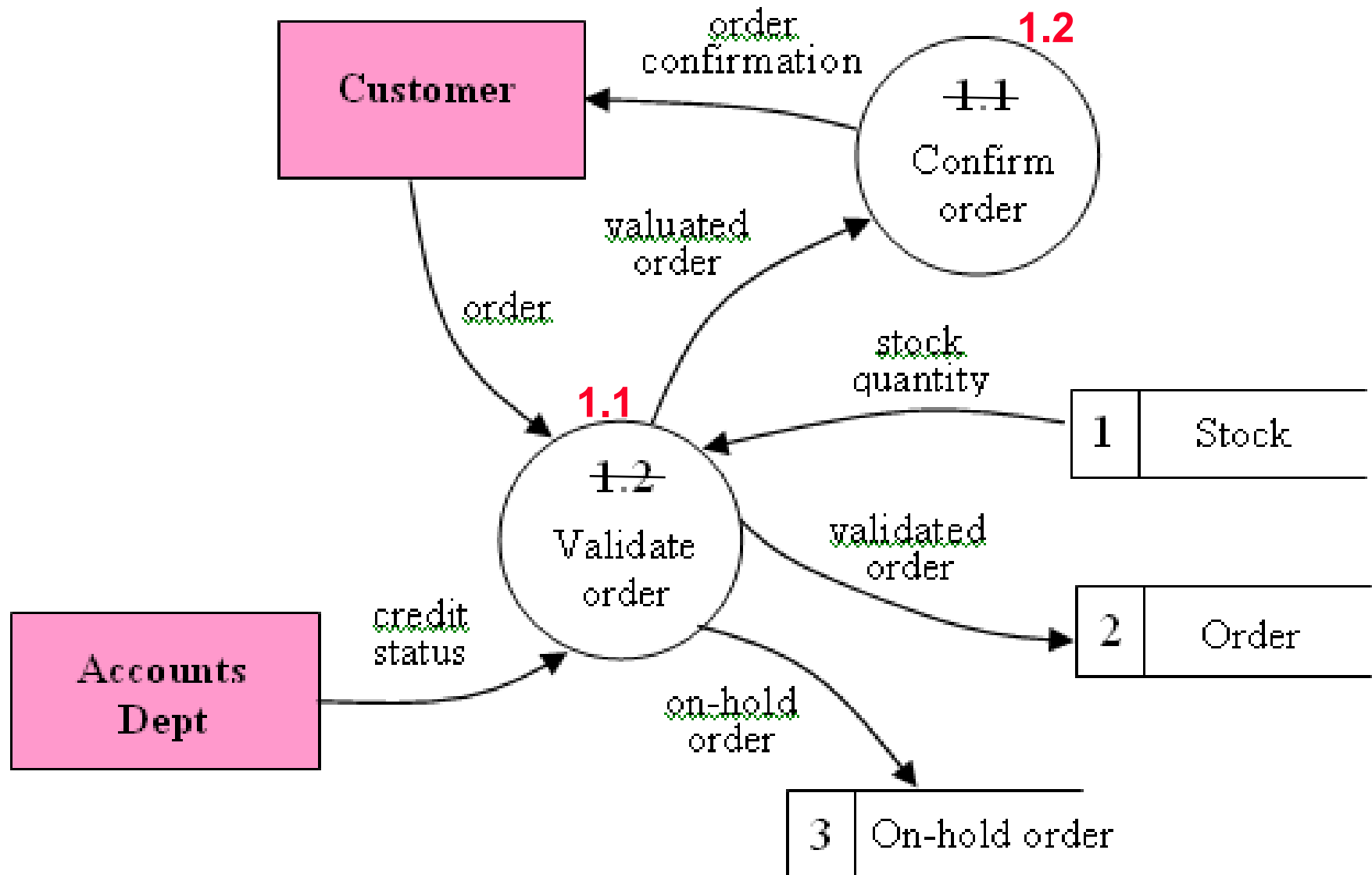
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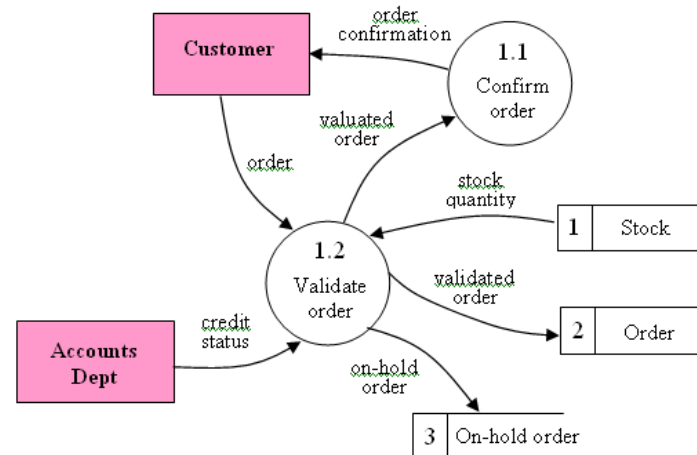


2. Data Flow Diagram

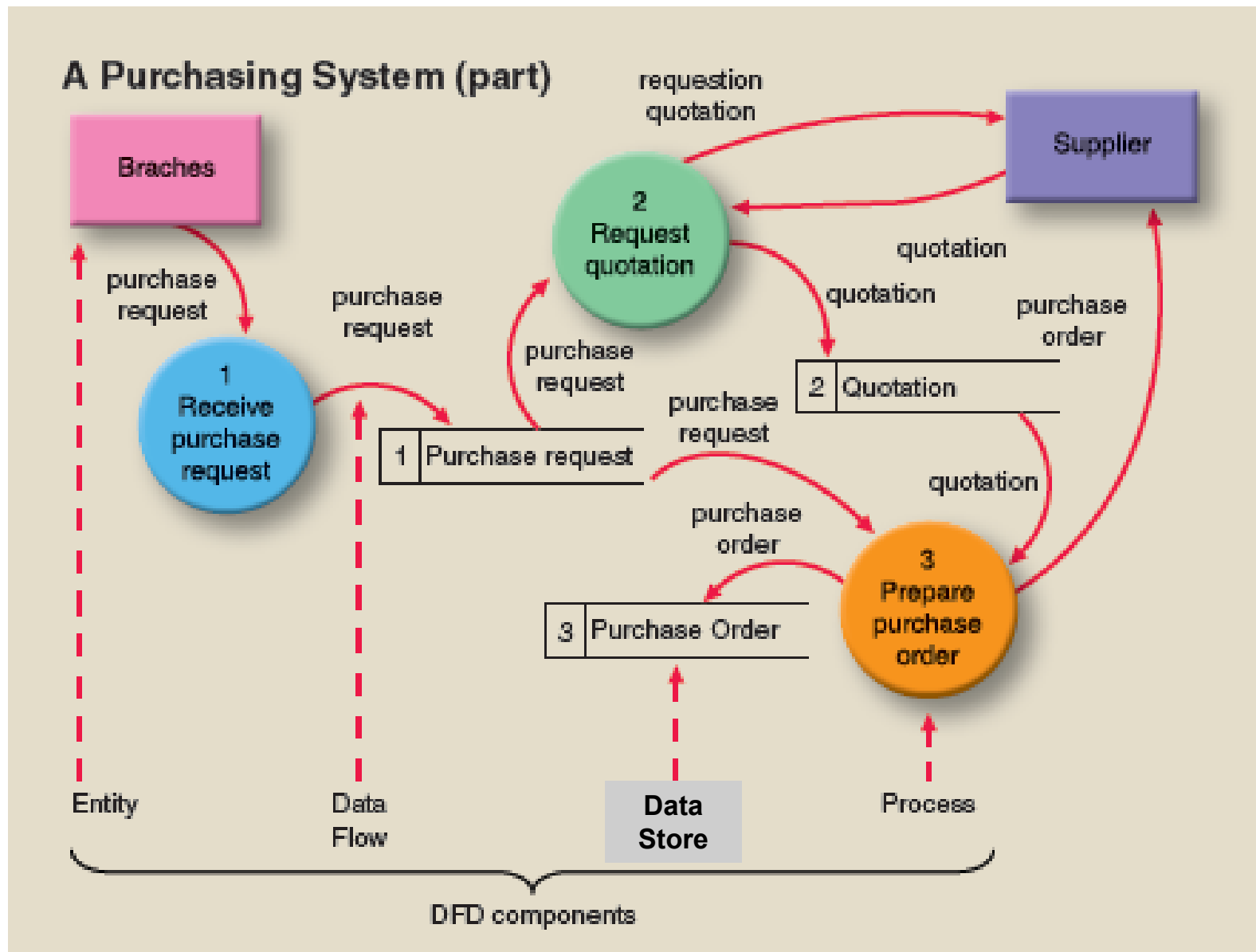


Definition of DFD

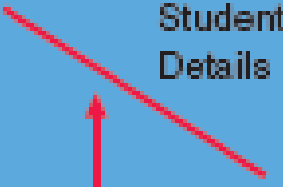
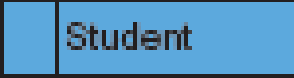
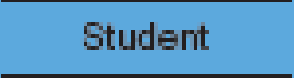
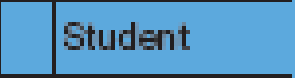
- **Graphical Models** – DFD uses graphical model to document the operation of the current system.
- **Processes** - The processes of the current system are modelled with a set of symbols or notations.
- **Notation/symbol** - The diagramming notation used: process, data store, data flow and entity.
- **Rules** - there are rules that govern the connections between these symbols



4 Components of DFD



4 Components of DFD (Popular DFD Symbols)

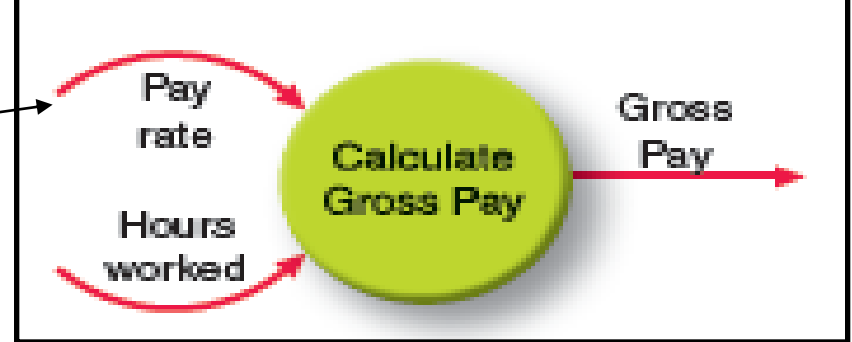
	de Marco	Gane & Sarson	Yourdon	Others
a. Data Flow				
b. Process				
c. Data Store				
d. Entity				

Notation Used (Recommended)

These will be the symbols recommended and used in **TAR UMT** in test, assignments and exams. system.

Components	Symbols	Description
Data Flow		A line with arrow head (can be straight or curve)
Process		A perfect circle
Data Store		An open ended rectangle
Entity		A rectangle

1. Data Flow



- **Definition**

- A **path** by which **data** moves from one part of the Information System (IS) to another part.
- Represents a specific piece of data in **motion**.

- **Examples**

- **Calculating Gross Pay Process** : Input data flows (eg. worked hours, pay rate, employee number); Output data flows (eg. gross pay)
- **Creating Purchase Order Process** : Input data flows (purchase requisition, item details, supplier details); Output data flows (eg. PO)

- **Symbol/Notation**



2. Process

- **Definition**

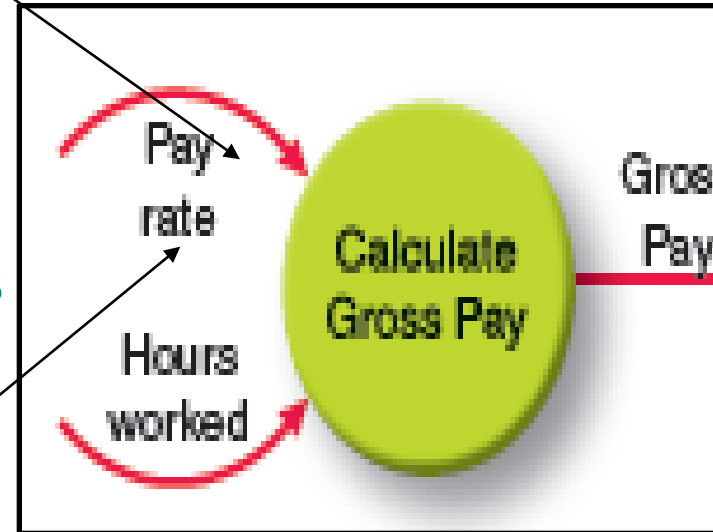
- Describes the main **transformations** that take place within the system. It changes incoming data flows into outgoing data flows.

- **Examples**

- Calculate gross pay
- Prepare vendor/supplier cheques

- **Symbol**

- A circle
- Labeled with an active **verb** (eg. Calculate) and a **noun**



3. Data Store

- **Definition**

- Shows information that is *stored* within the system
- It is accessed and updated by the processes
- It is a **file or table**.

- **Examples**

- Student Scores
- Employee Salary and Deduction data



- **Symbol/Notation**

- An open-ended rectangle
- Labeled with a name (**noun**) and a number as an identifier

4. Entities

- **Definition**

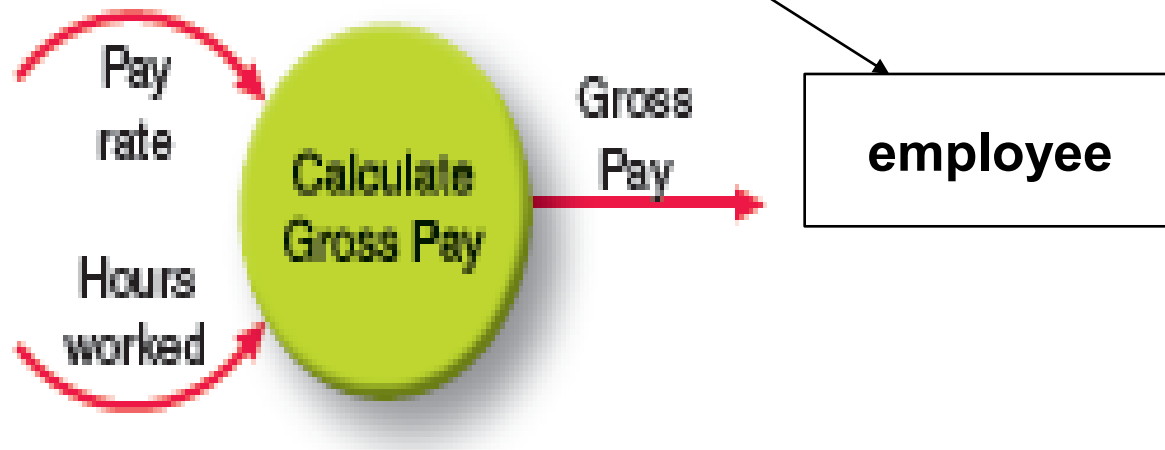
- Represents a person, department, organisation or other IS that send (source) or receives (sink) information to/from the **system** concerned.
- Show the **boundaries** of the IS

- **Examples**

- Customer
- Patient

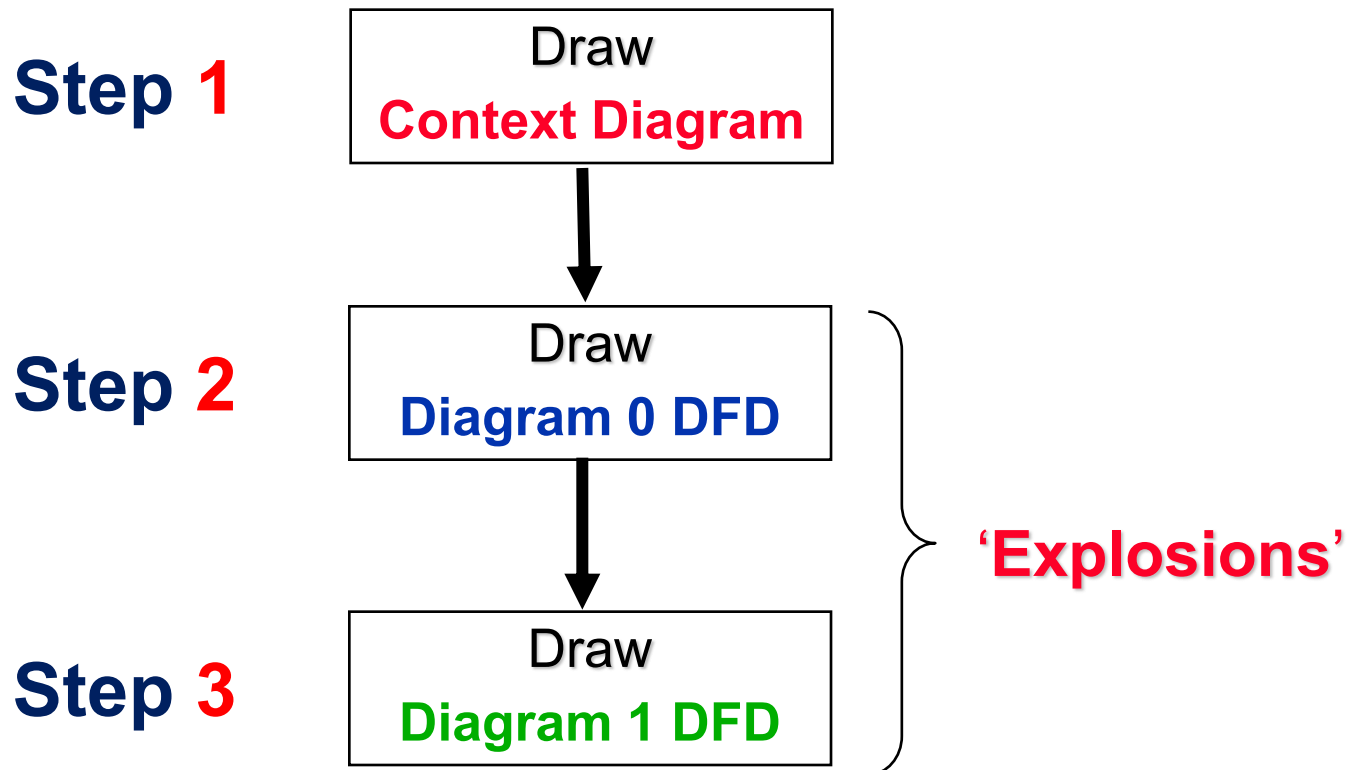
- **Symbol**

- A rectangle
- Labeled with a **noun**



Steps in Drawing DFD

- Start by drawing a **Context Diagram**
- Draw detail diagrams using series of '**explosion**' or level of abstraction eg. Diagram 0, diagram 1 etc. until '*functional primitive*'.

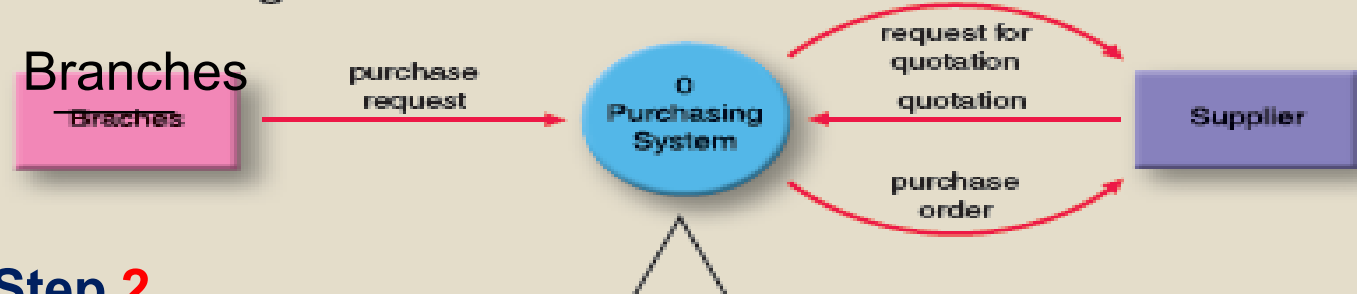


DFD and Levels of Abstraction ('explosion')

Step 1

Context Diagram

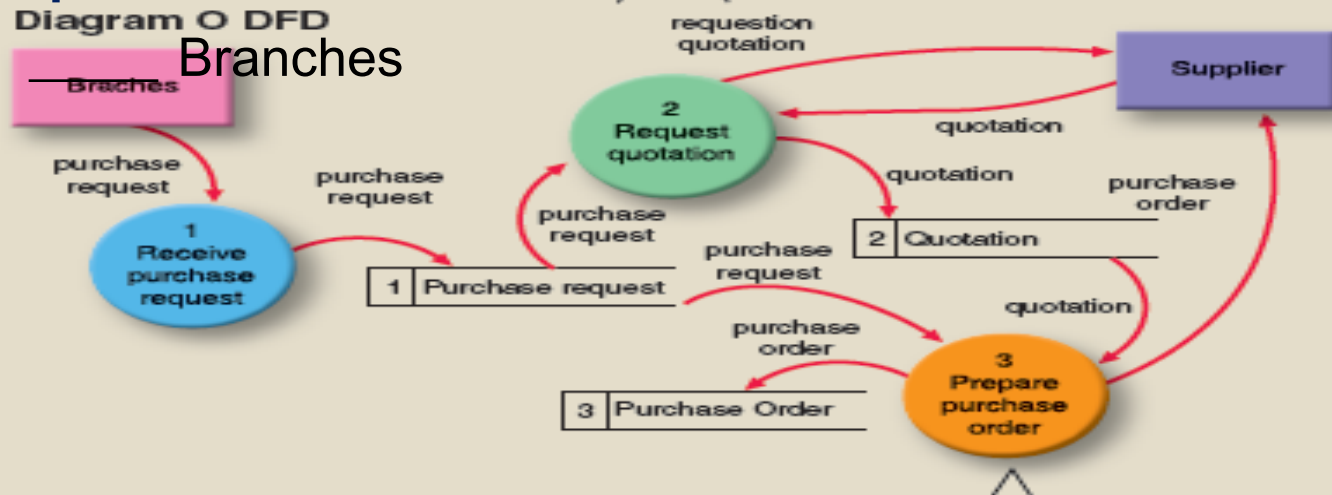
Branches



Step 2

Diagram 0 DFD

Branches



Step 3

Diagram 1 DFD



Note :

1. Data stores not shown in context diagram
2. The number of external entities remain the same
3. In 'balancing', the number of inflows and outflows remain the same

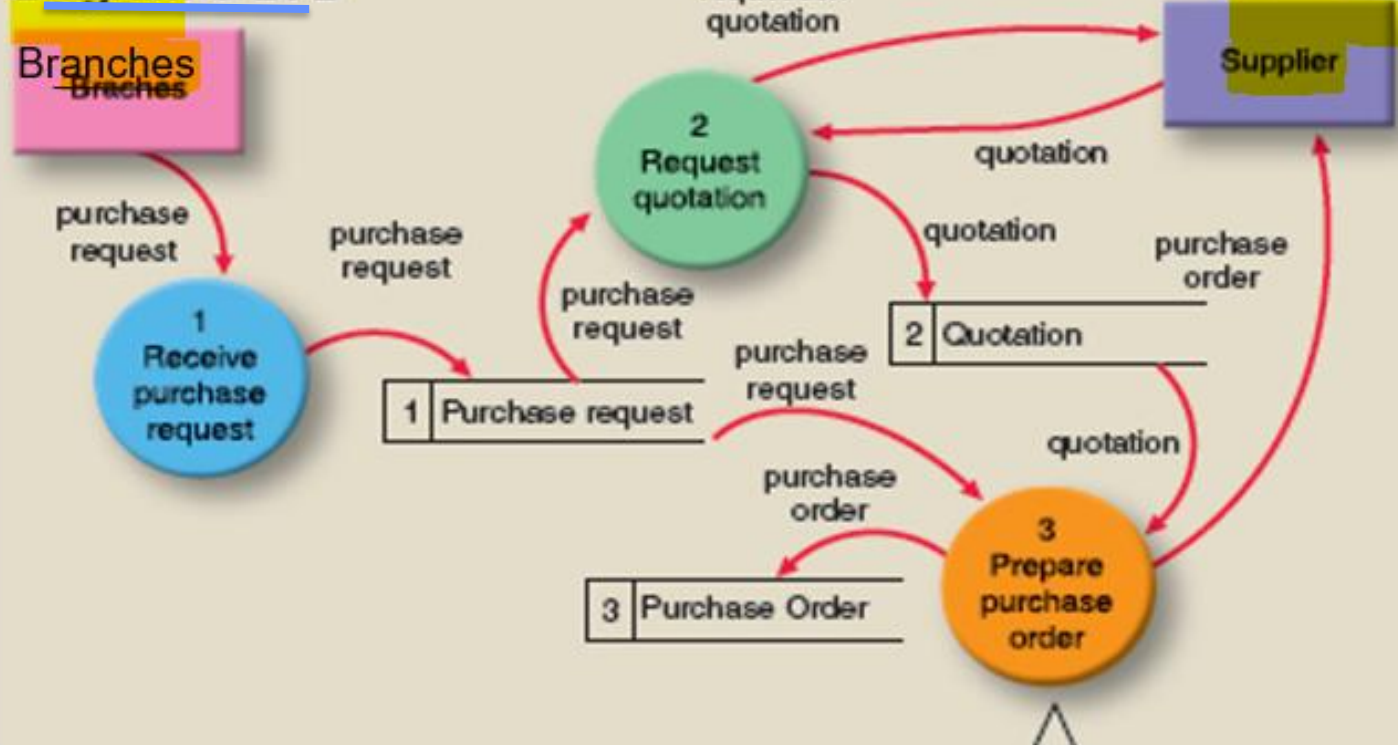
Step 1

Context Diagram



Step 2

Diagram O DFD



Note :

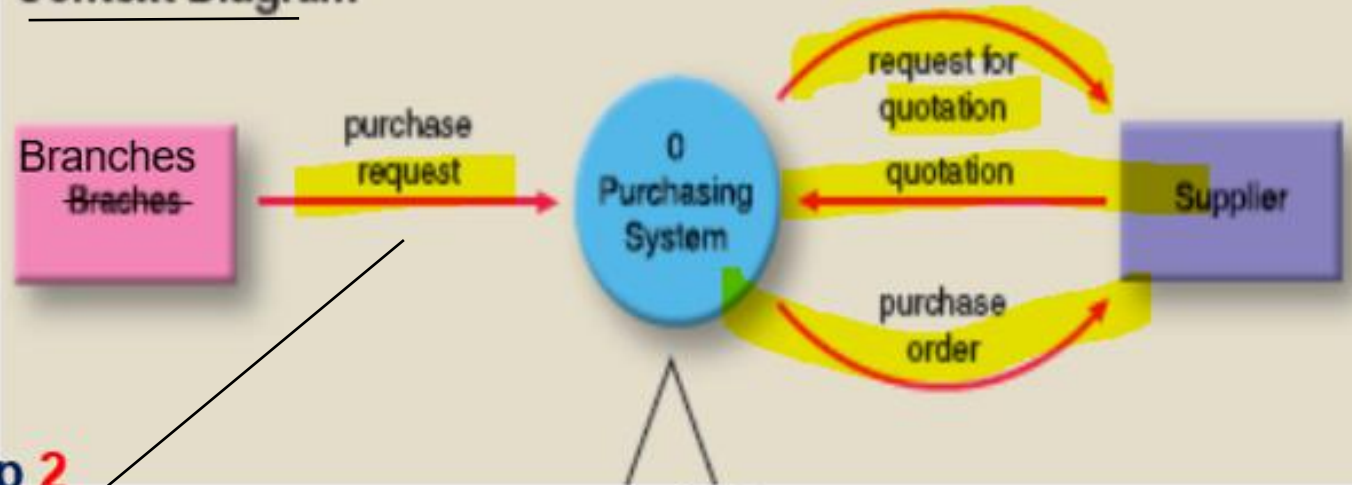
1. Data stores not shown in context diagram
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Step 1

Note :

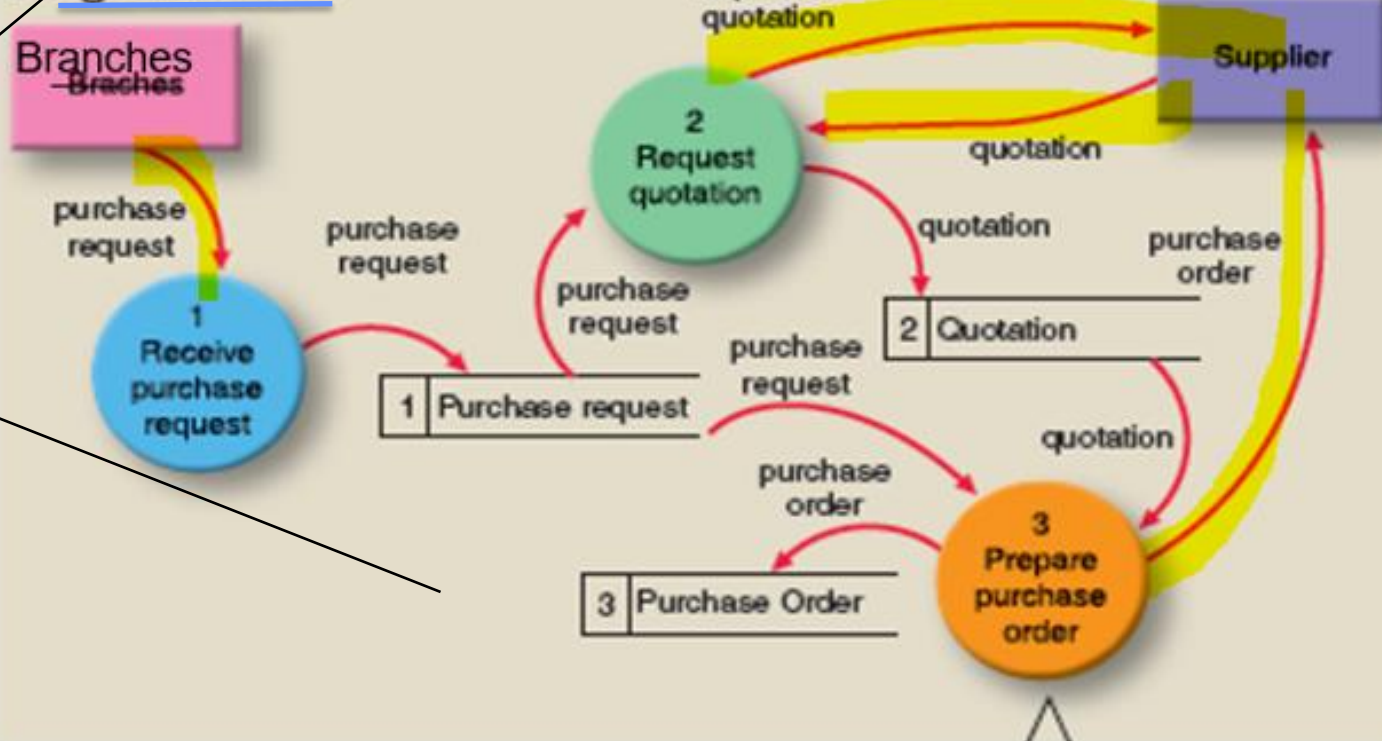
1. Data stores not shown in context diagram
2. The number of external entities remain the same
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Context Diagram



Step 2

Diagram O DFD

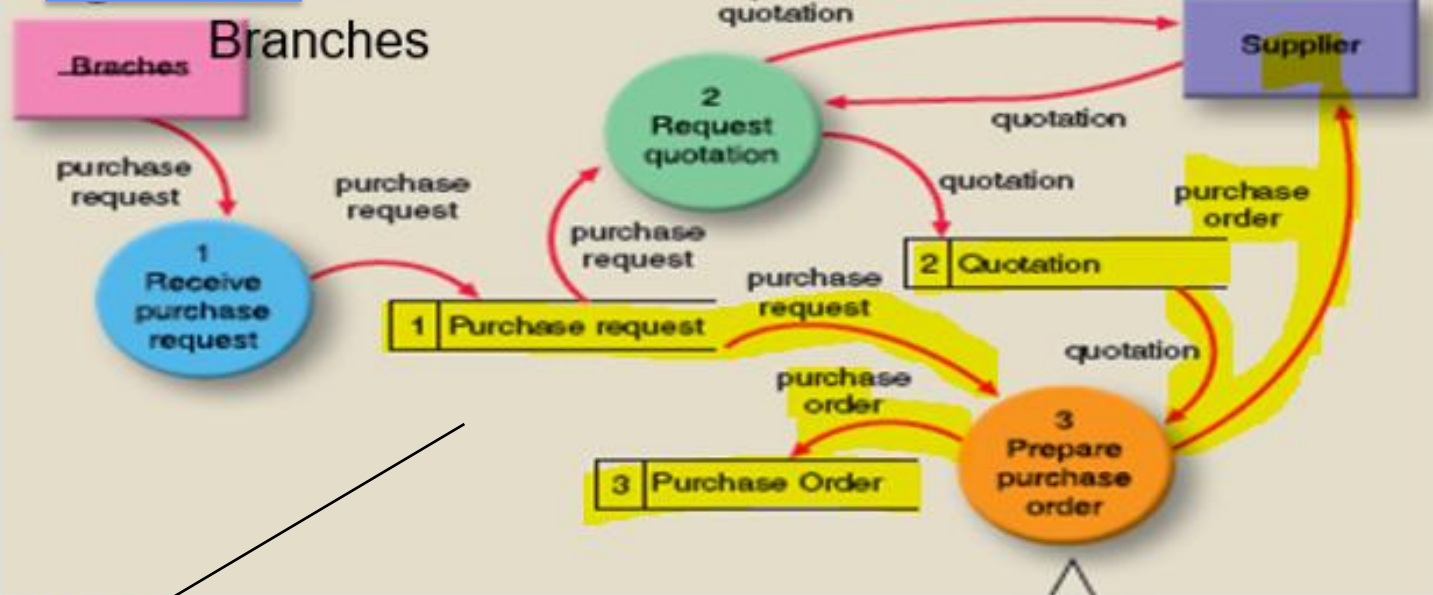


Step 2

Note :

1. Data stores not shown in context diagram
2. The number of external entities remain the same
3. In 'balancing', the number of inflows and outflows remain the same

Diagram 0 DFD



Step 3

Diagram 1 DFD





Purchase request

Last Monday -Perak - Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack accept
Last Thurday -Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack accept

Purchase Request File /data store

PR01	-Last Monday -Perak Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack	Susan	Last Tuesday	delivered-close case	PO 1
PR02	-Last Thurday - Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack	Susan	Last Thurday	delivered-close case	PO 1

Quotation

Request Quotation

RQ1 -Susan -Last Friday - Supplier A - want buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price ?
RQ2 -Susan -Last Friday - Supplier B - want buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price ?



Quotation

Quotation File /Data Store

Last Friday - Supplier A	- want buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price RM2000	Susan	Last Friday
This Monday -Supplier B	- want buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price RM3000	Susan	This Monday

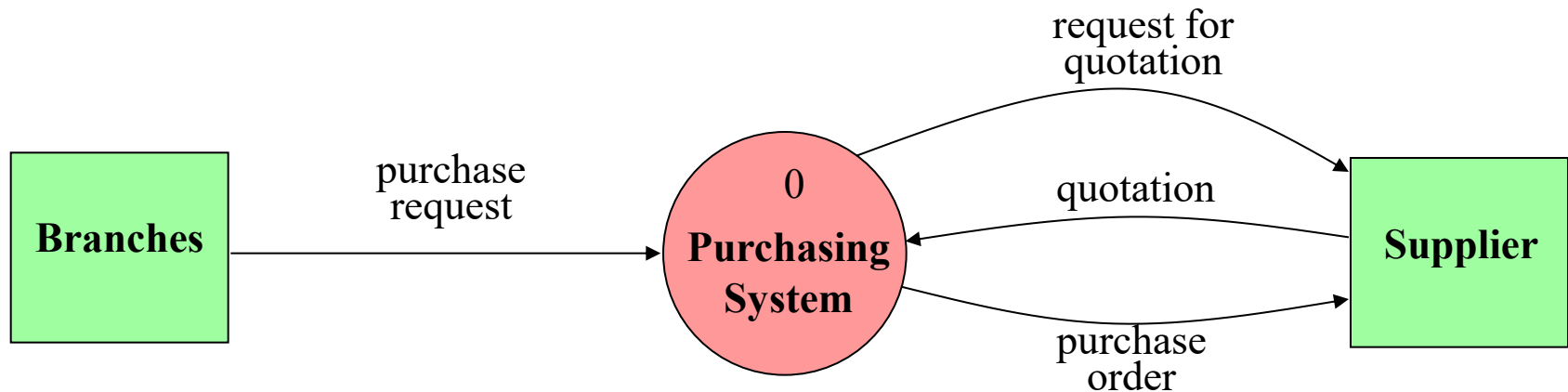
Double a Paper A4 70g 500's
PAPER ONE A4 70gsm...

Purchase Order File /Data Store

PO1 -This Tuesday Susan Supplier A buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price RM2000

Step 1 - Context Diagram

- Definition** – represents the entire software element as a **single process** (circle) with input and output data indicated by incoming and outgoing arrows. Boundary indicated by external entities.



Example 1 – Sales Order System

- In XYZ Ltd, when a sales order is **received** from a **customer**, the sales personnel would have to **check** whether the particular customer's credit limit has been exceeded or not. This can be carried out by checking with the **Accounts Dept**. The next thing to check is the availability of the stock items ordered.
- Where all these are in proper order, then an order confirmation will be **sent** to the customer and a **delivery request** sent to the **Warehouse**. The Warehouse would **confirm** if the delivery of goods has been completed. After the goods have been sent to the customer, an Invoice generation **request** will be sent to the Accounts Dept to issue an invoice to the customer.

invoice

LOGO

FROM

East Asia Trading
Pasar Pudu Baru 10
Kuala Lumpur 53000

INVOICE

MY-001

INVOICE DATE

29/01/2019

P.O.#

2330/2019

DUE DATE

24/05/2019

BILL TO

Mayang Bujang
2 Pasar Moden 55
Kuala Lumpur 55100

SHIP TO

Mayang Bujang
1721 Jln Sp 232/7
Kuala Lumpur 55103

QTY	DESCRIPTION	UNIT PRICE	AMOUNT
1	Wooden elephant figurine	600.00	600.00
2	Large cloth rice bag	45.00	90.00
3	Bamboo ladder	20.00	60.00

Subtotal 750.00

SST 6.0% 45.00

TOTAL

RM795.00



TERMS & CONDITIONS

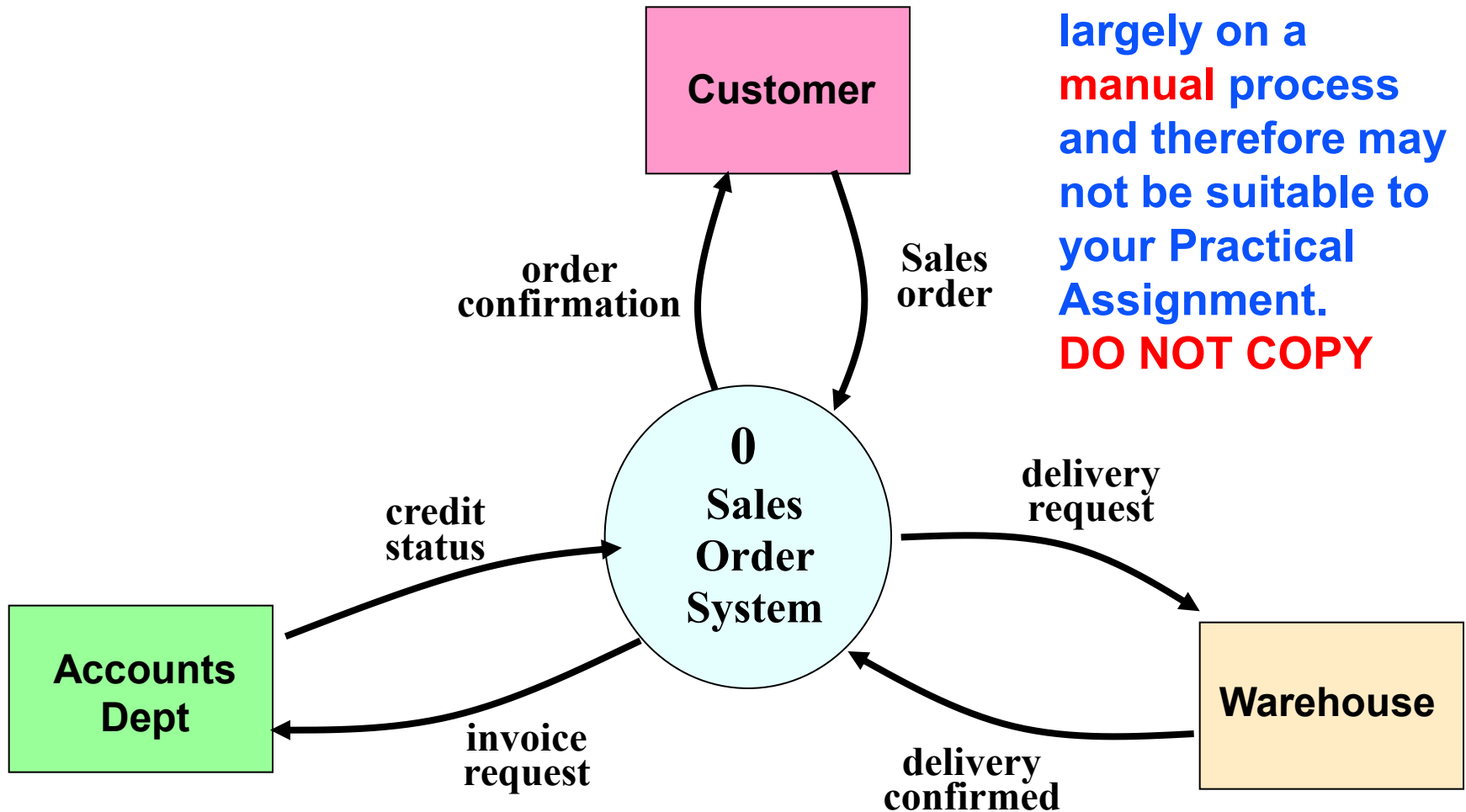
Payment is due within 15 days

Public Bank Berhad
Account Number: 12345678
Routing Number: 0987654321098

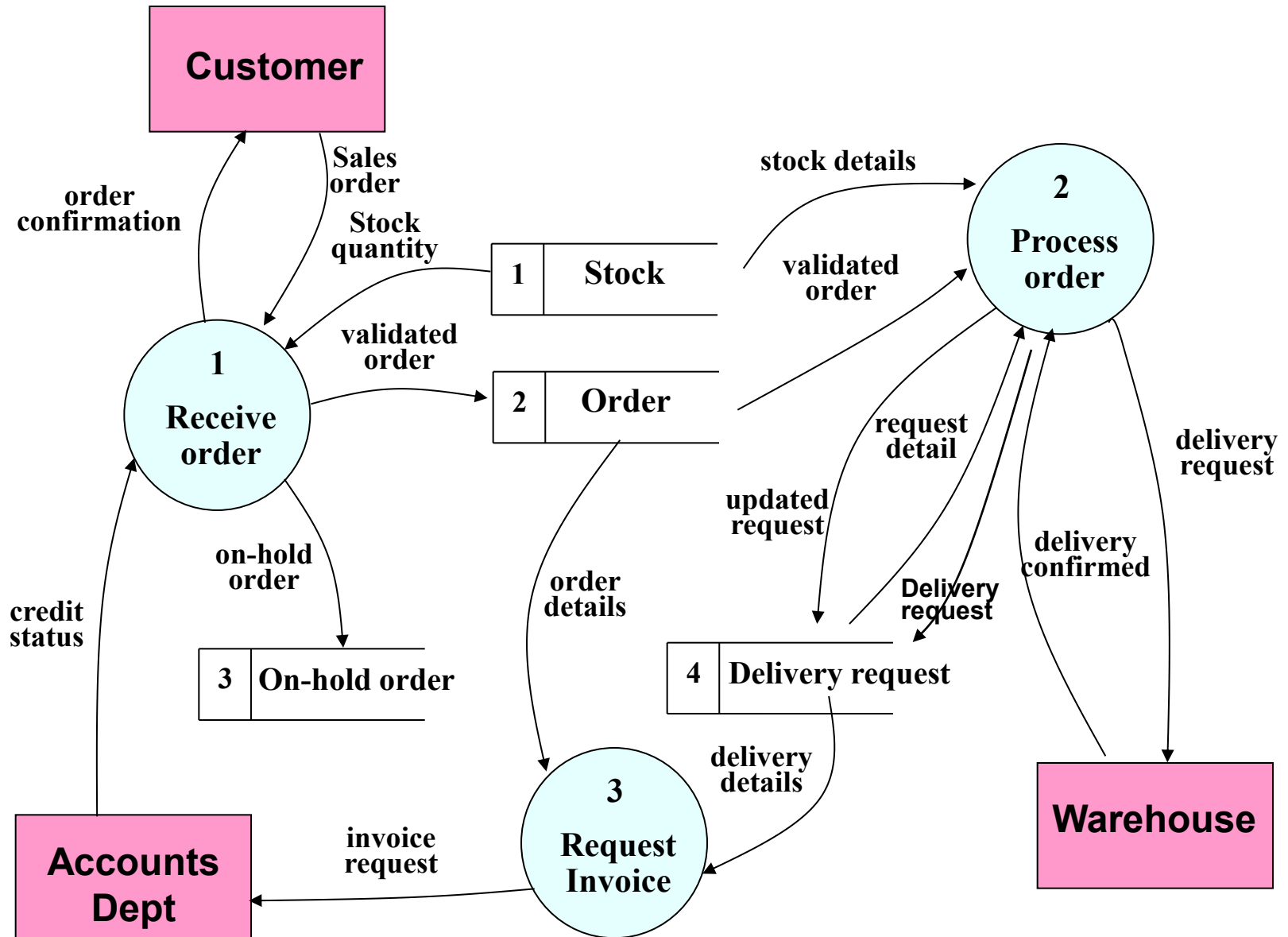
<https://invoicehome.com/>

Context Diagram

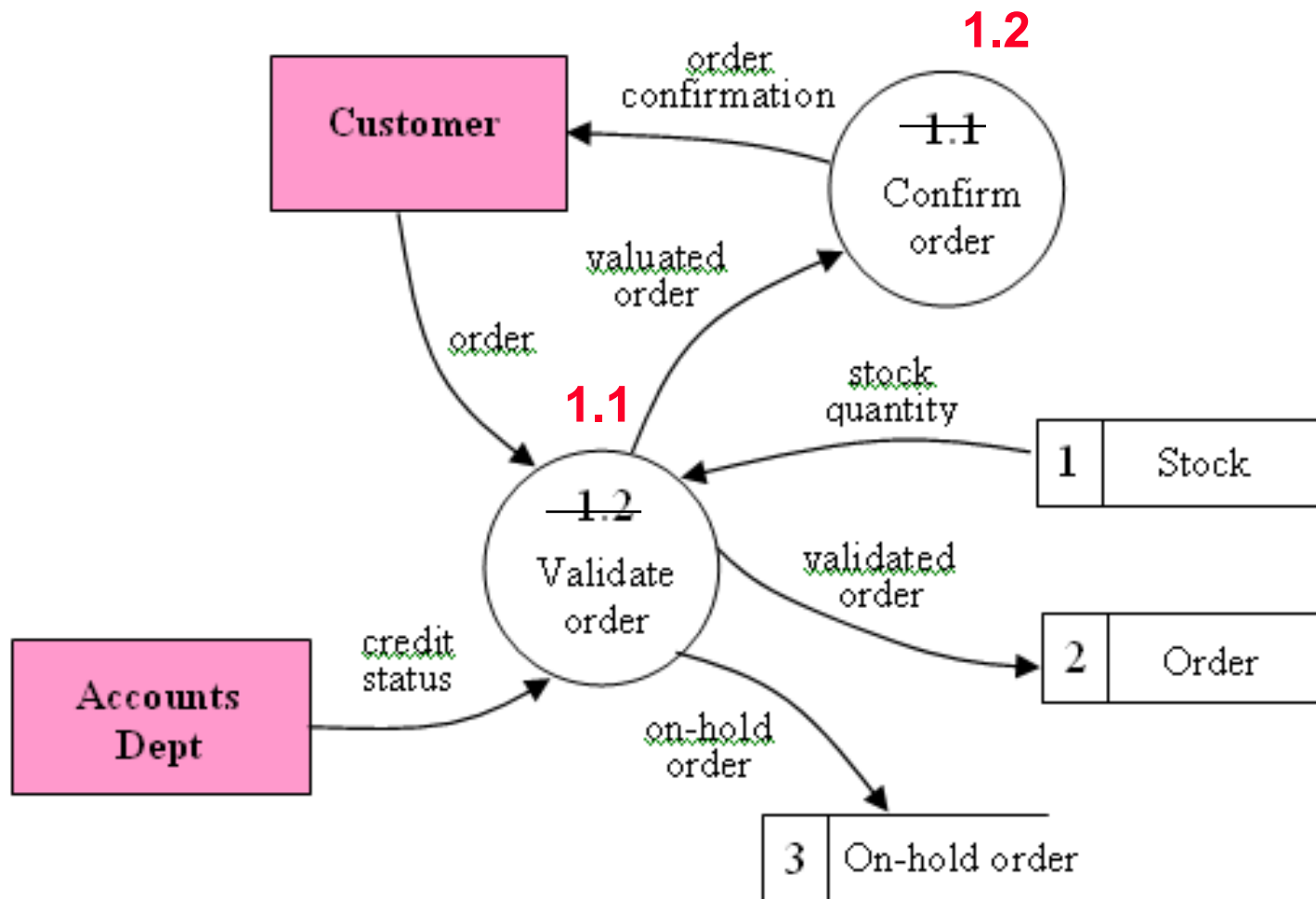
Note : that this system has been drawn based largely on a **manual** process and therefore may not be suitable to your Practical Assignment.
DO NOT COPY



Step 2 - Diagram 0 DFD



Step 3 - Diagram 1 DFD (for Receive Order process)



Example 2 – Ticket Reservation System

- The ticketing function starts with the **customer booking** tickets online in the comfort of their homes or simply to walk-in to any of the 50 ticketing counters located in various places throughout the country. When a reservation is made, cash **payment** is collected to be **banked** in later in the day. Payment can also be made using credit cards where the normal credit verification with the **credit card firm** is carried out. **Cancellation**, if any, can only be allowed at least two weeks prior to the **concert** when **refunds** will be given. In other cases no refunds will be allowed.

Context Diagram – Ticket Reservation System

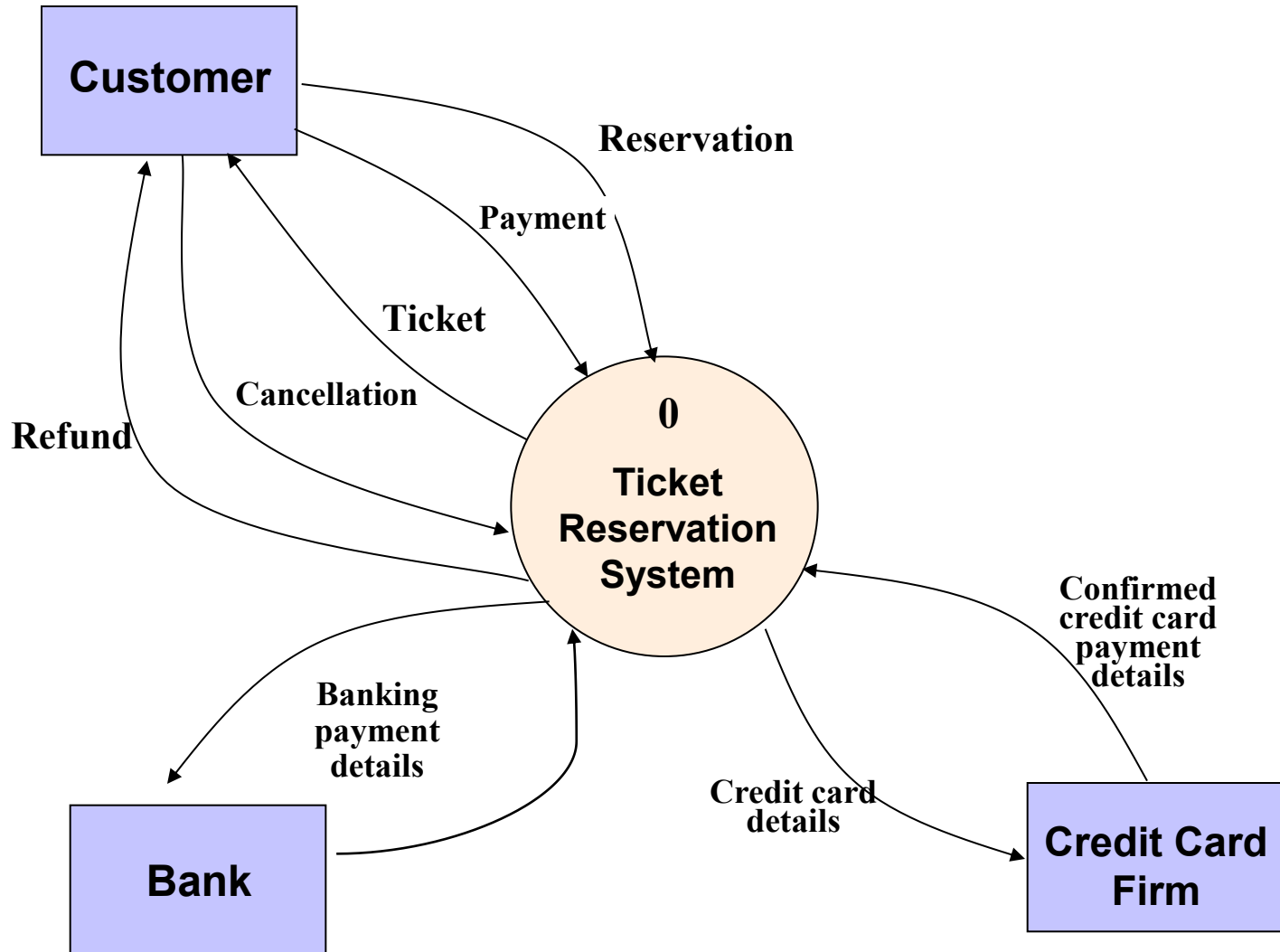
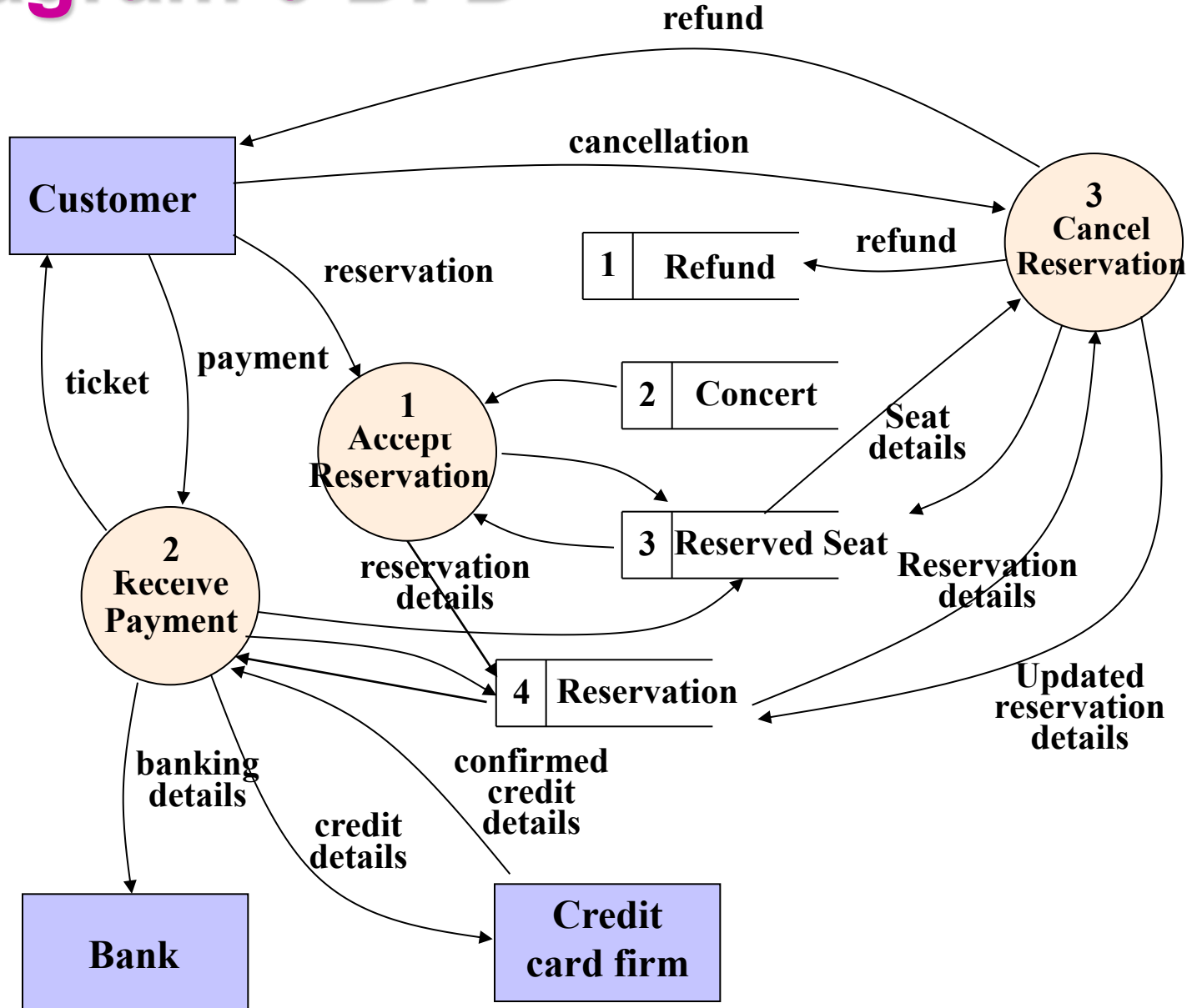


Diagram 0 DFD



Steps in Drawing DFD

- Purchasing System

Example 3 - Purchasing System

- Purchase requests are **received** from other **branches** to buy goods. The company practices a centralized purchasing policy.

Purchase request	
Last Monday -Perak - Ali request to buy 10 boxes - A4 papers (500 sheets per package), 1 box -4 pack	accept
Last Thursday -Johor - David request to buy 20 boxes - A4 papers (500 sheets per package), 1 box -4 pack	accept

- The purchasing department will **invite** request for quotations to **suppliers**. Suppliers will be **sending** in quotation for biddings.

Request	Quotation
RQ1 -Susan -Last Friday - Supplier A - want buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price ?	
RQ2 -Susan -Last Friday - Supplier B - want buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price ?	

Quotation	Quotation	Selected quotation
Last Friday -Supplier A - want buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price RM2000		
This Monday -Supplier B - want buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price RM3000		

rejected quotation

Example 3 - Purchasing System

- A purchase order will be **prepared**, a copy of which is sent to the supplier and a copy retained. Occasionally, new suppliers may be included into the suppliers file.

Purchase Order	File /Data Store														
PO1 -This Tuesday	Susan	Supplier A	buy 30 boxes, A4 papers (500 sheets per package), 1 box -4 pack -- Price RM2000												

- When goods are delivered by the suppliers, the items **received** are checked against the purchase order. The delivery order is filed after noting any discrepancies.

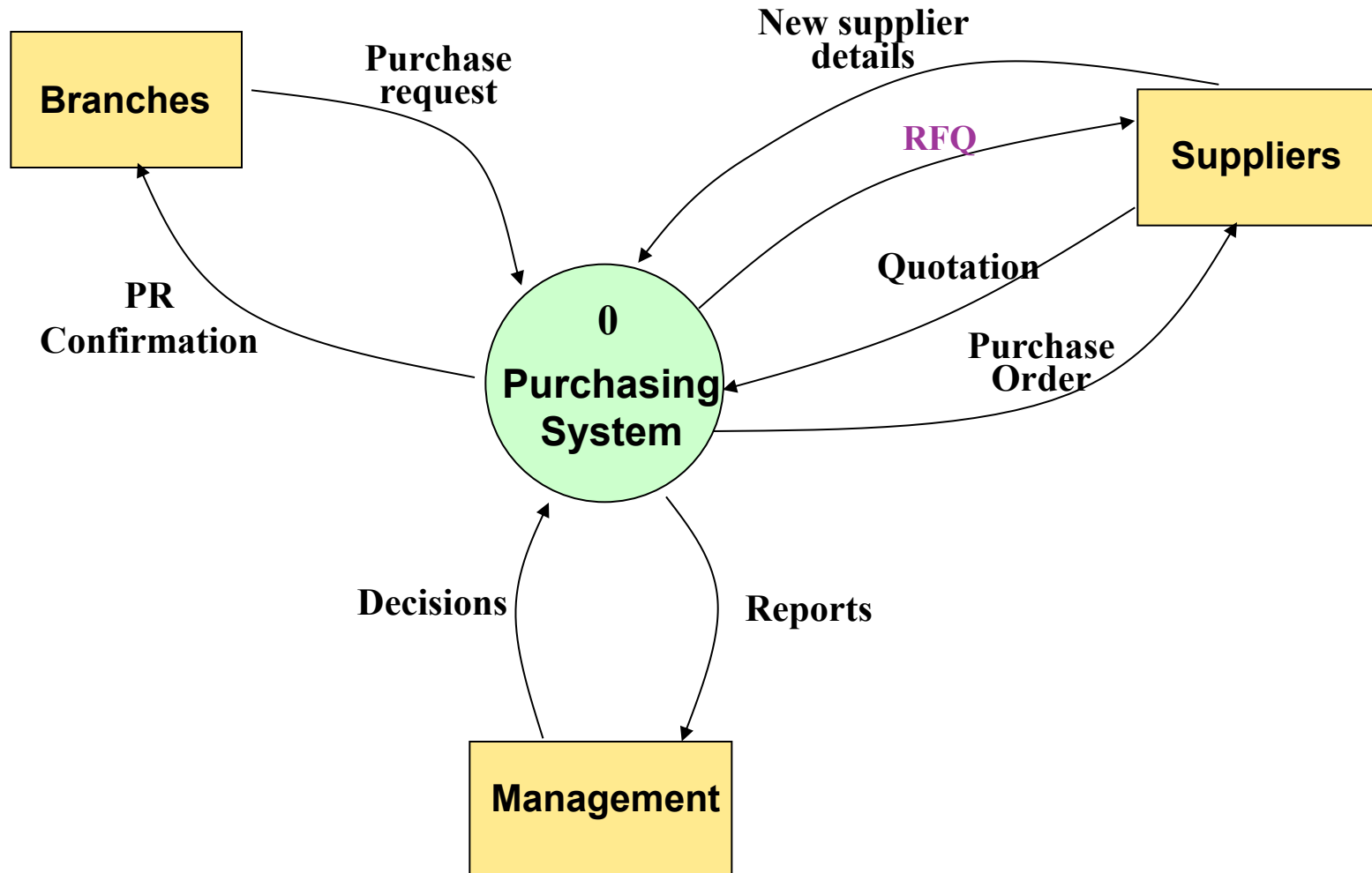
(ordered 30boxes, but supplier delivered 20 boxes)

- When the invoice is received from the supplier, it is checked against the delivery order, payment is then **made** and the invoice is filed.

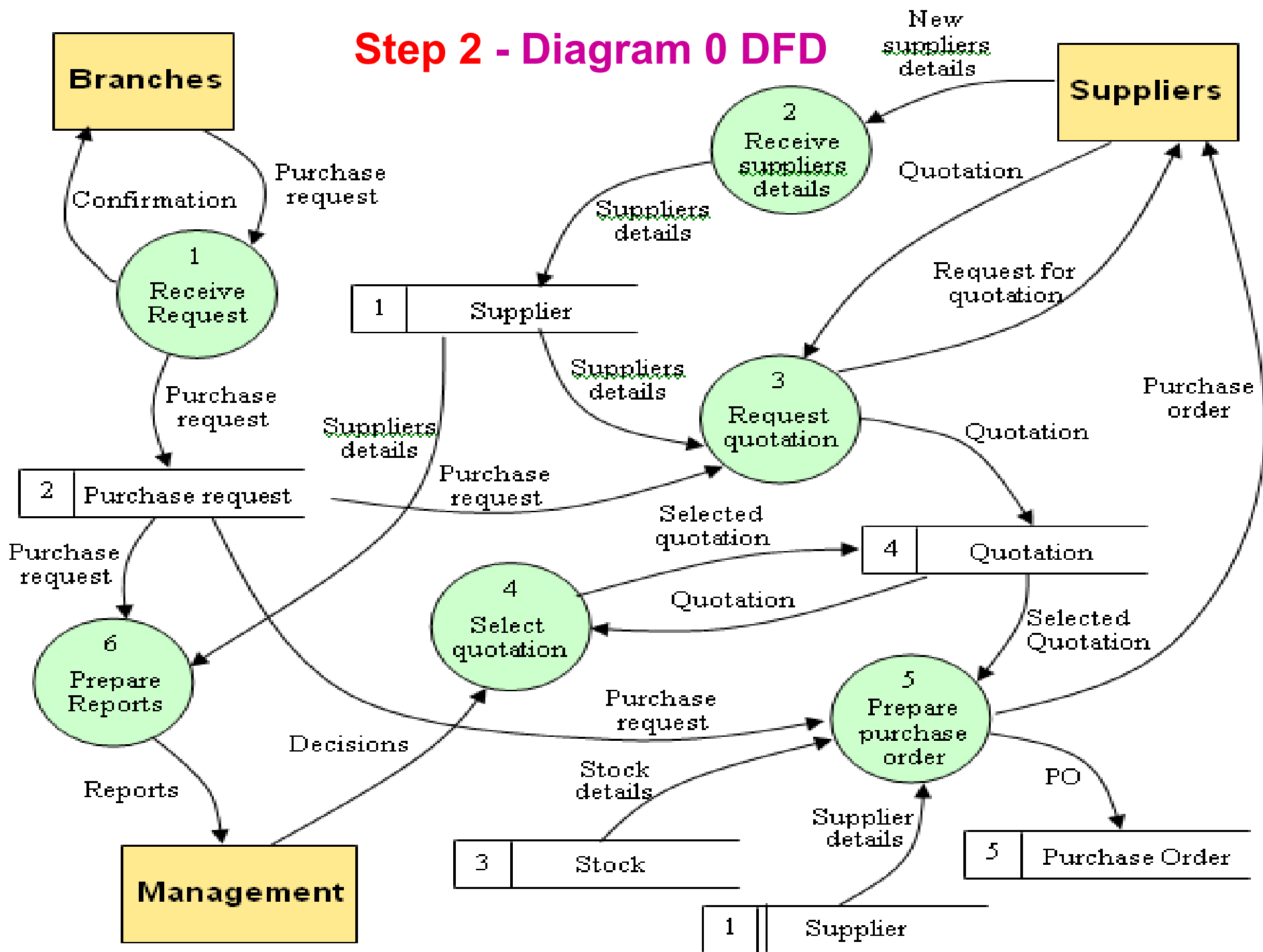
Step 1- Draw Context Diagram

- Draw a **single process** that represents the entire system. The process is **named** after the system with **number zero**
- Identify and draw the external **entities**
- Draw the **data flows** from an external entity to the system and from the system to that external entity
- **NOTE** : logical **assumptions** on data flows may be made for this system. Other data flows which are not within the **functions** of a purchasing system will have to be left out.

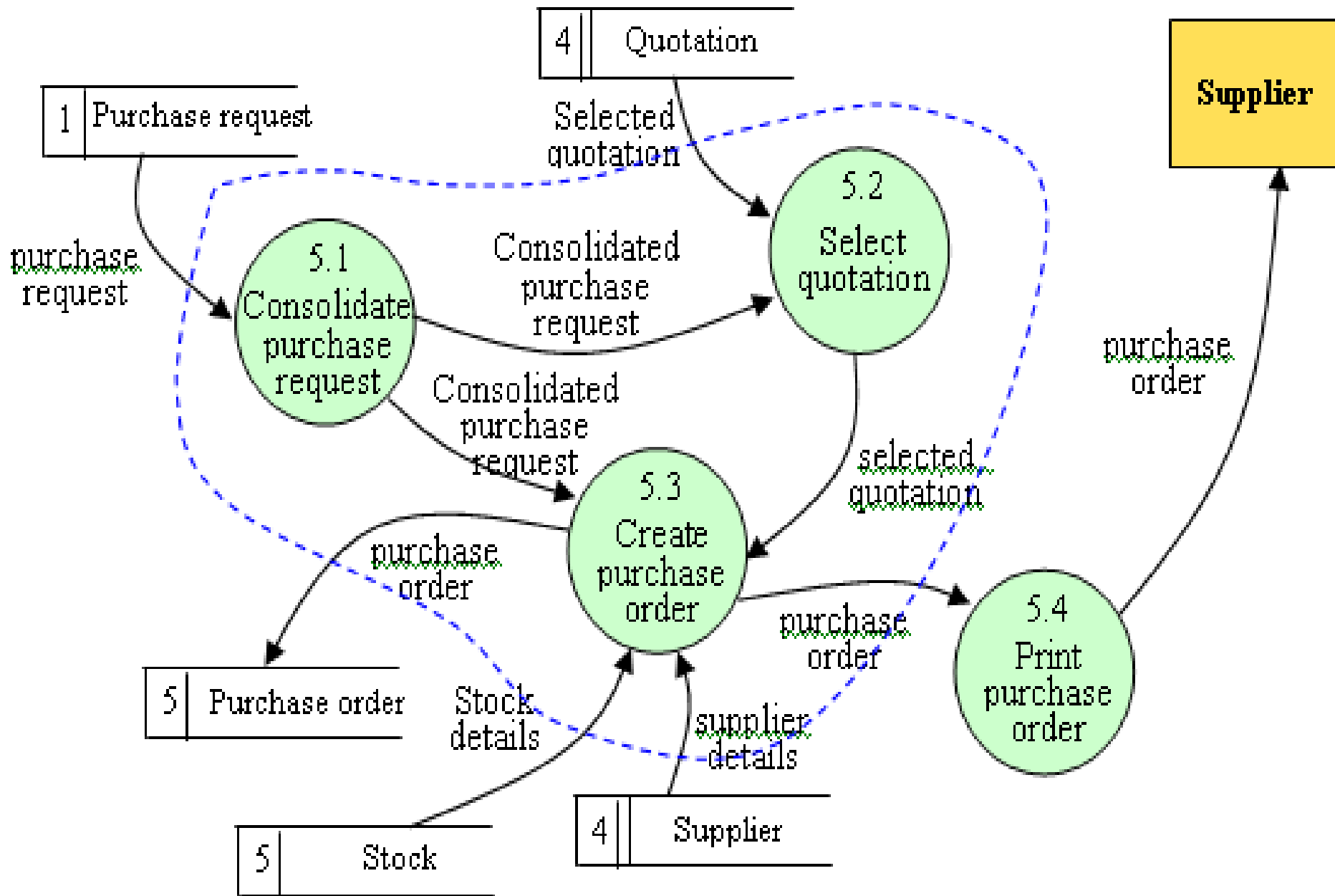
Context Diagram



Step 2 - Diagram 0 DFD

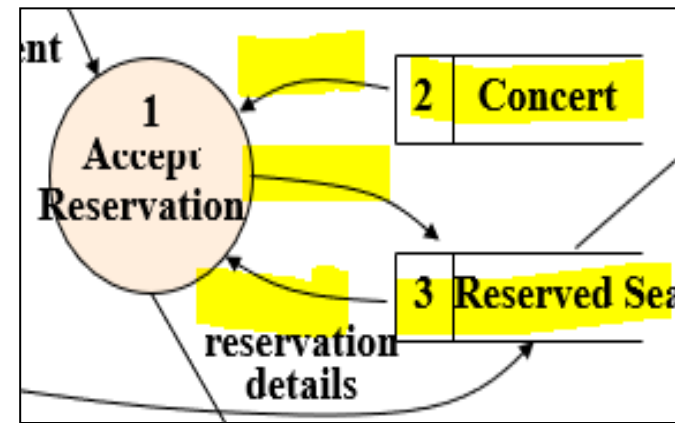
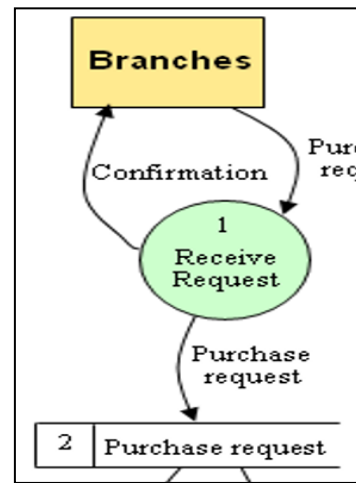
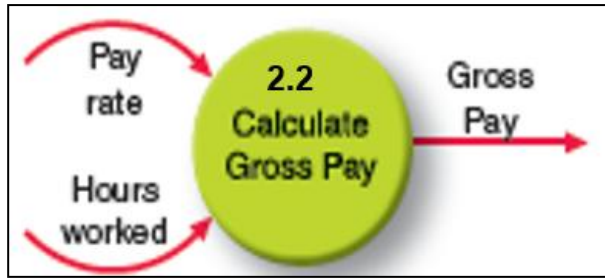


Step 3 - Diagram 1 DFD (for Prepare PO process)



Important Points (Rules on DFD)

DFD Rules



1. Process Rules

- **Transformation.** Processes must have both **inputs** and **outputs** and they must differ, that is some transformation must occur to the inputs to produce the outputs
- **Process Number.** Each process is numbered but this is only for ease of identification and not to show sequence.
- **Process Heading.** The description of the process should be a **verb** like 'prepare', 'calculate', 'check' etc plus **objects** (nouns).

2. Data Flow Rules

- **Arrows.** Arrows must always finish at or start from a process.
- **Label.** Every arrows must be labeled. However, there is no requirement to label arrows that go in and out of data stores.

3. Data Store Rules

- **Number.** Data stores are given a reference number (for reference only)

4. Repeated Diagrams Rules

- Rather than having data flow arrows criss-crossing all over the place it is often simpler to show a symbol more than once on the same diagram.

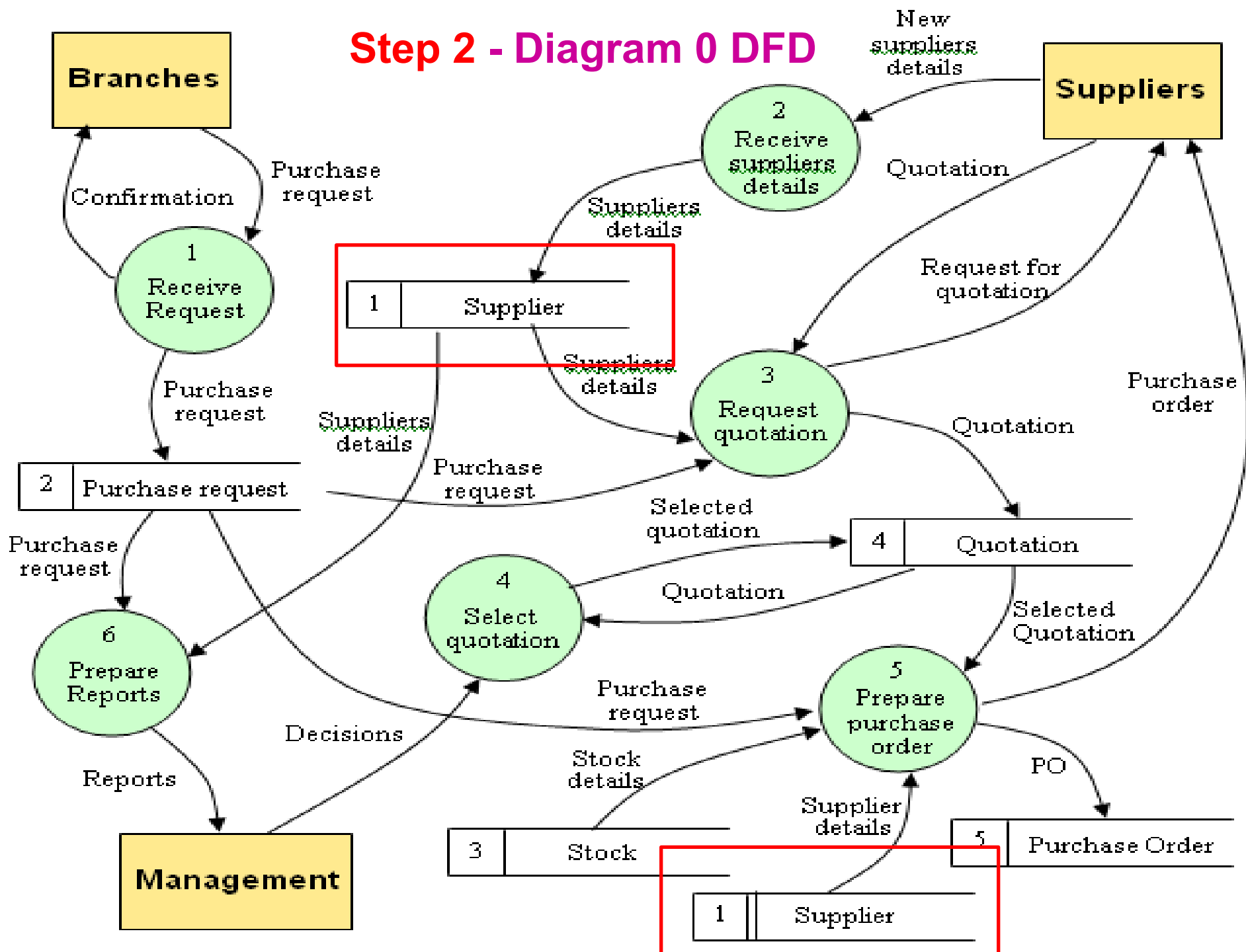
- Repeated Data Store



- Repeated Entity



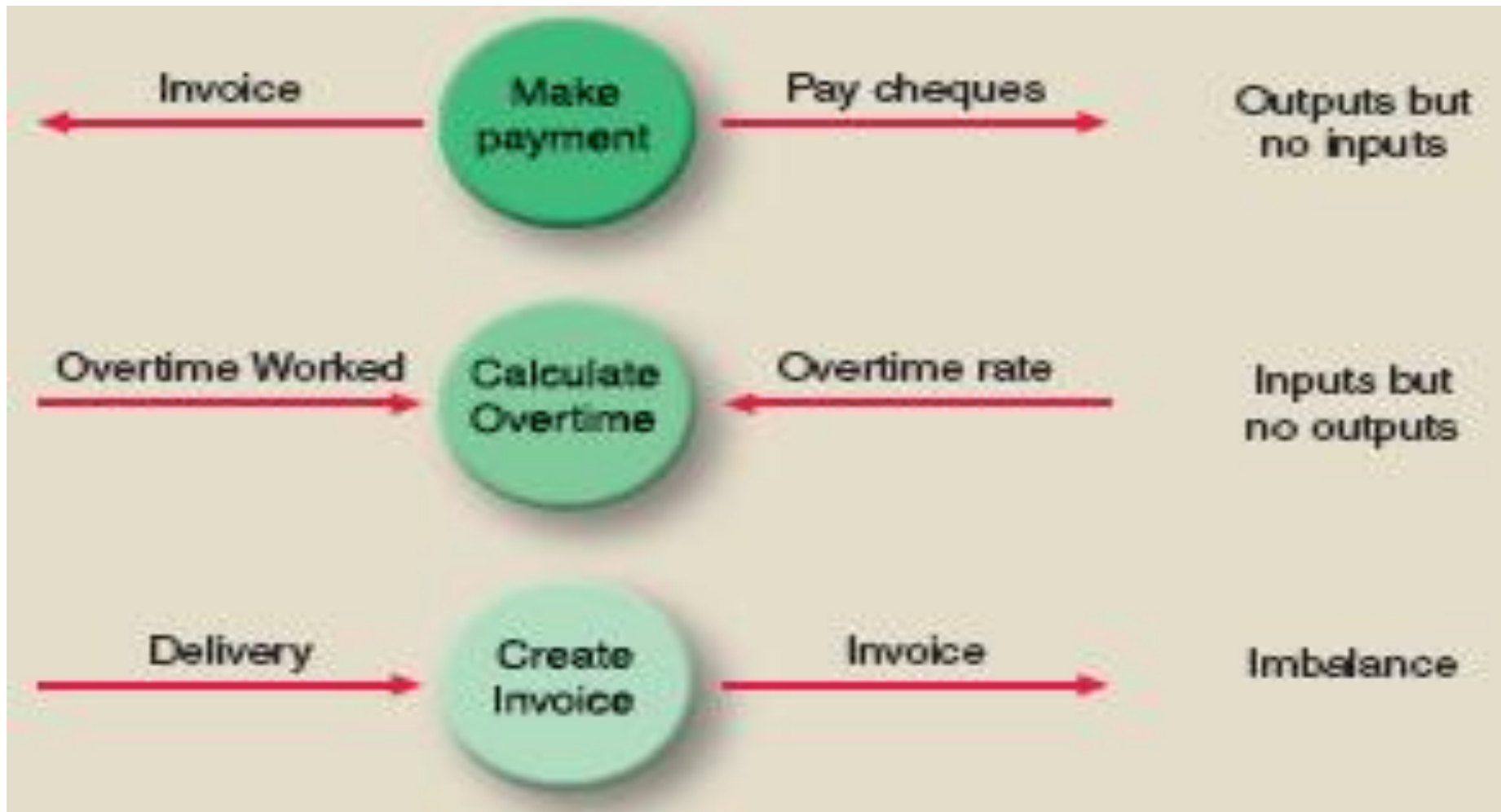
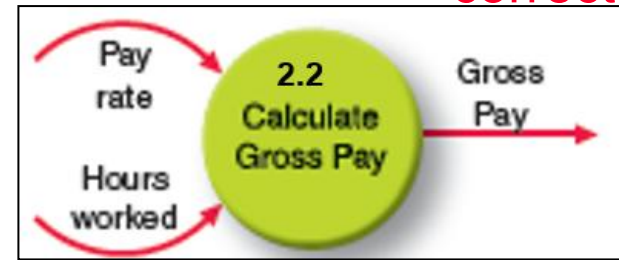
Step 2 - Diagram 0 DFD



Some Common Errors

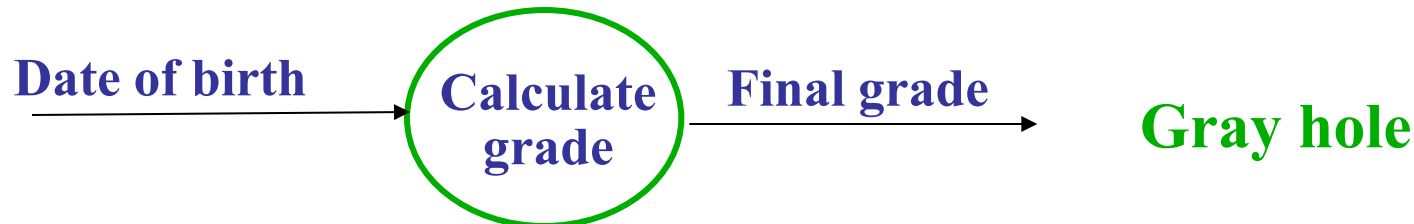
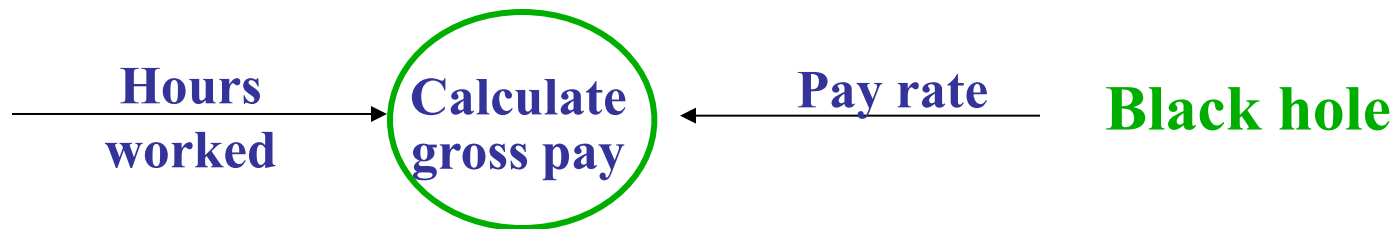
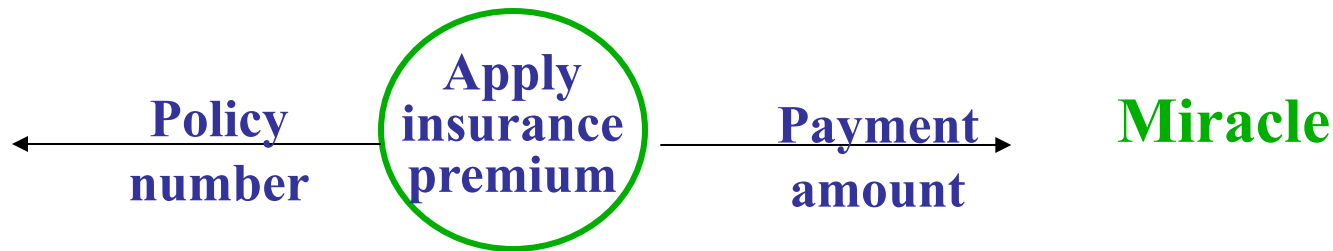
Incorrect Combination of Data Flow and Process Symbols

correct

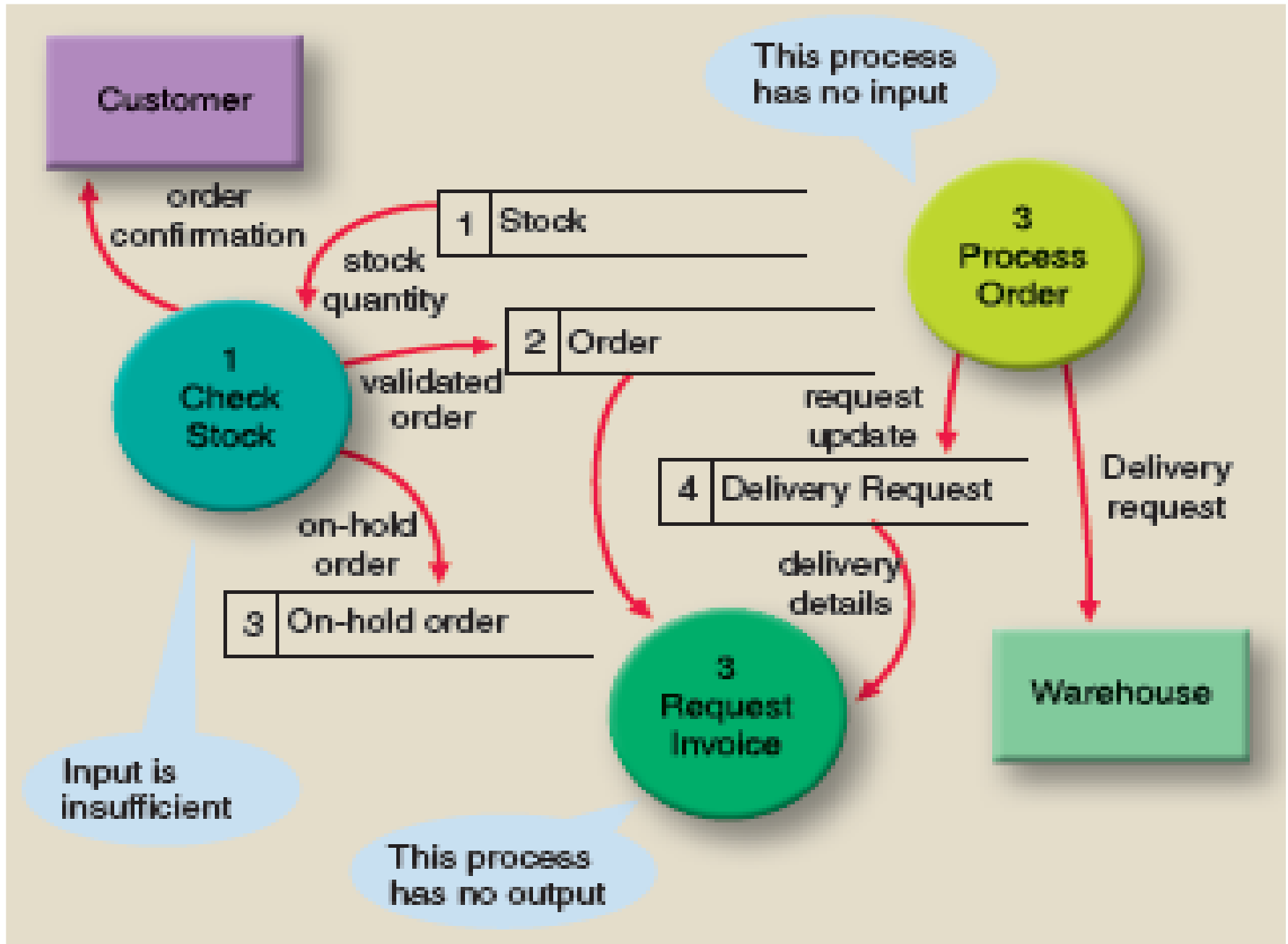


Incorrect Combination of Data Flow and Process Symbols

- **Miracle**: has outputs but no inputs
- **Black hole**: has inputs but no outputs
- **Gray hole**: the inputs are insufficient to produce the output



Example of Common Errors



Correct Combination of Data Flow and Process Symbols



Incorrect Uses of Data Store Symbols



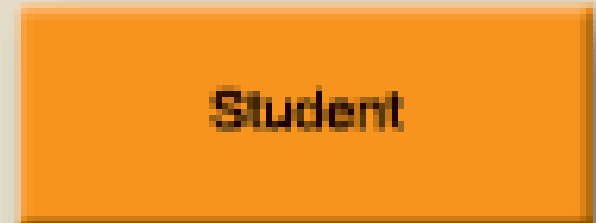
Results



No intervening
process present

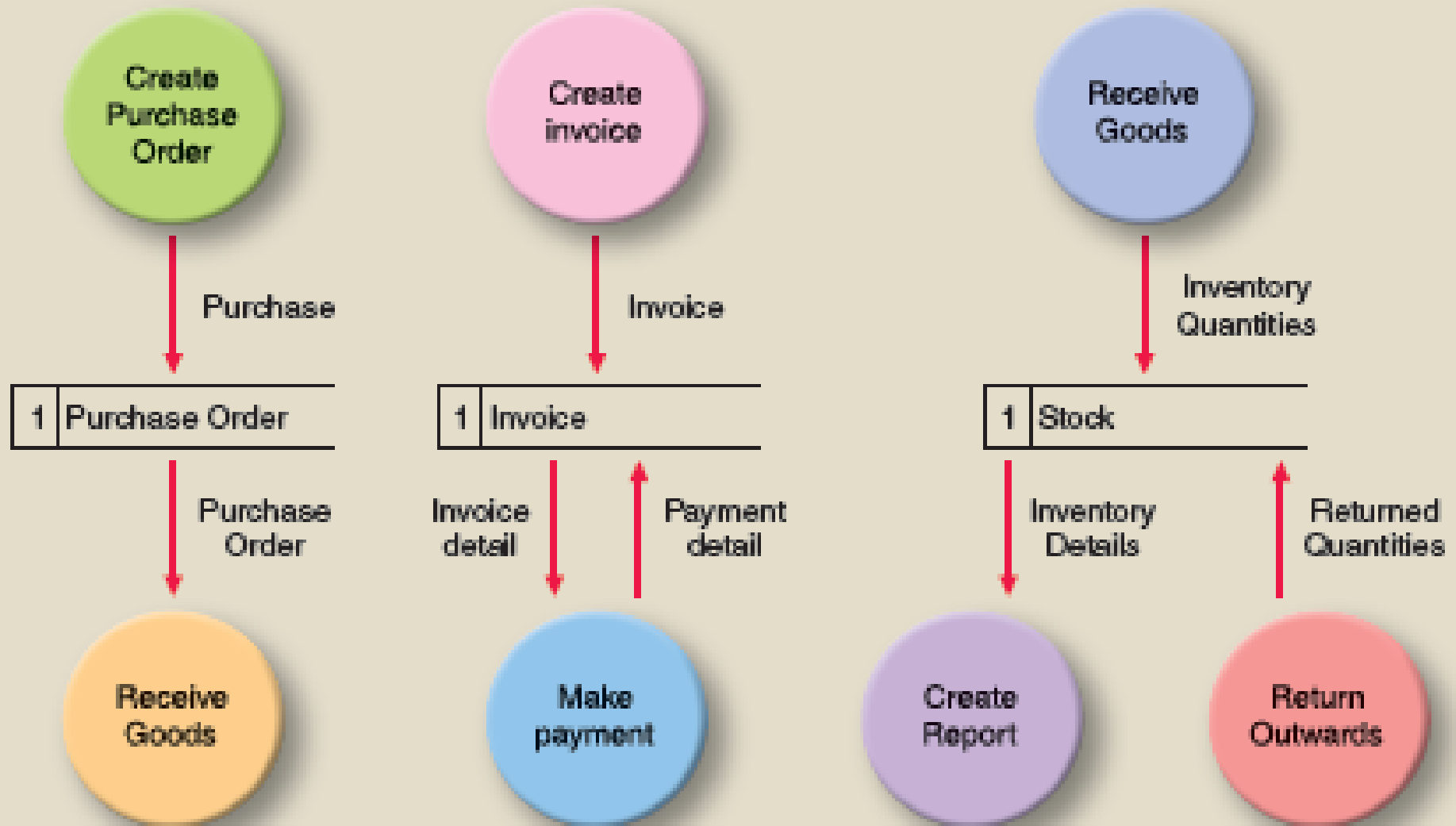


Course
details

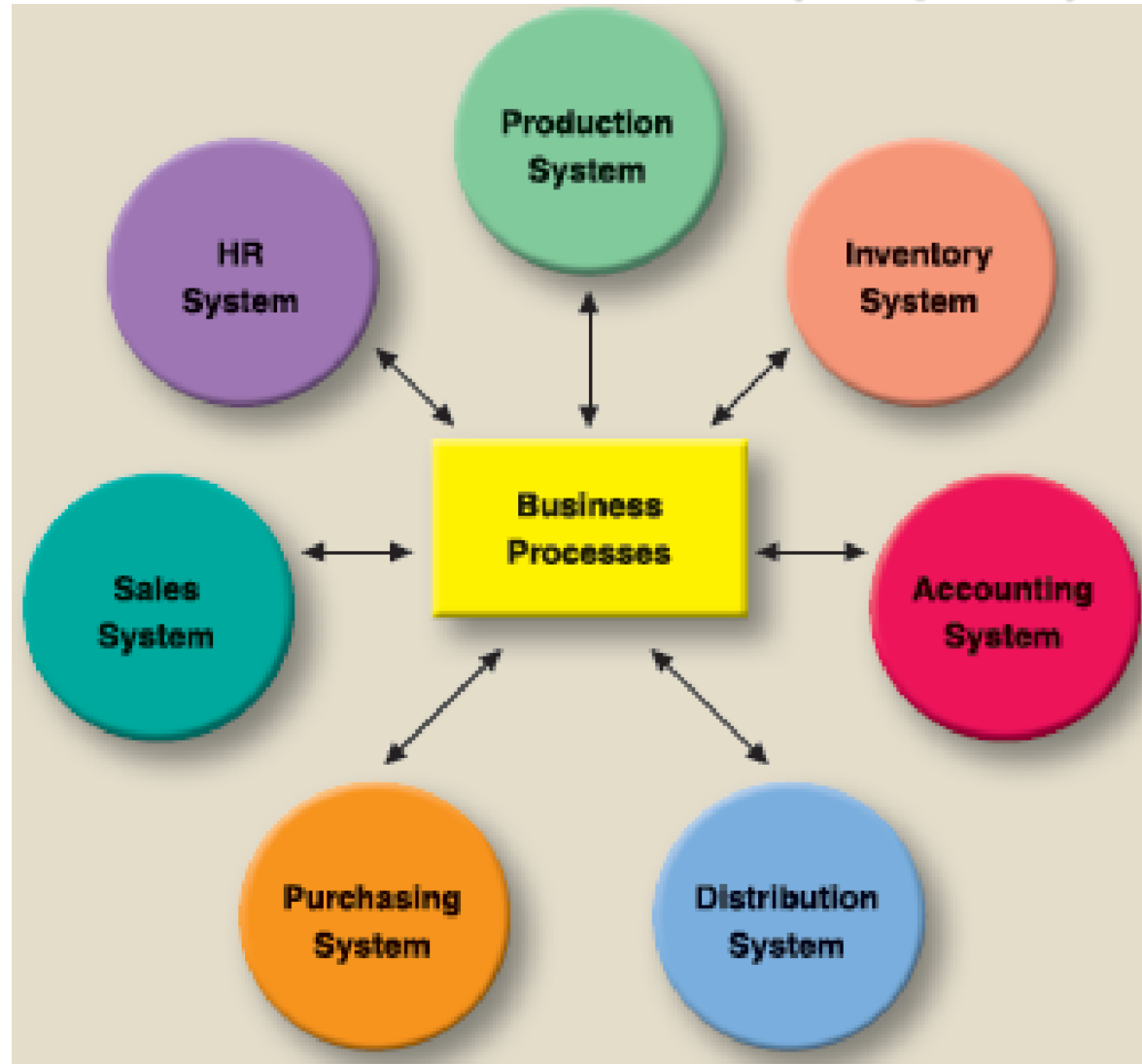


No intervening
process present

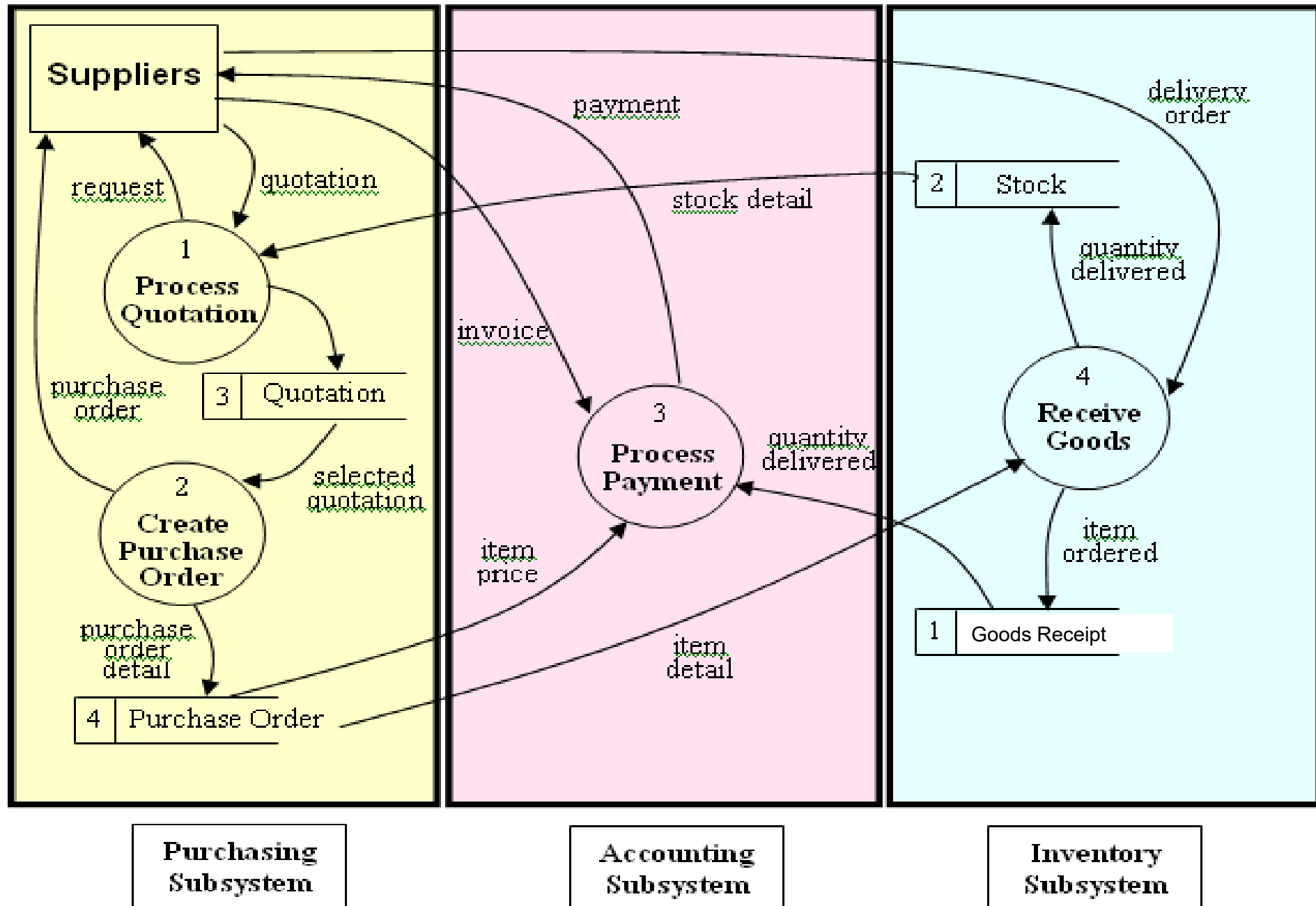
Correct Use of Data Store Symbols



- **Enterprise Resource Planning (ERP) system**
Integrated Processes in ERP (chapter1)



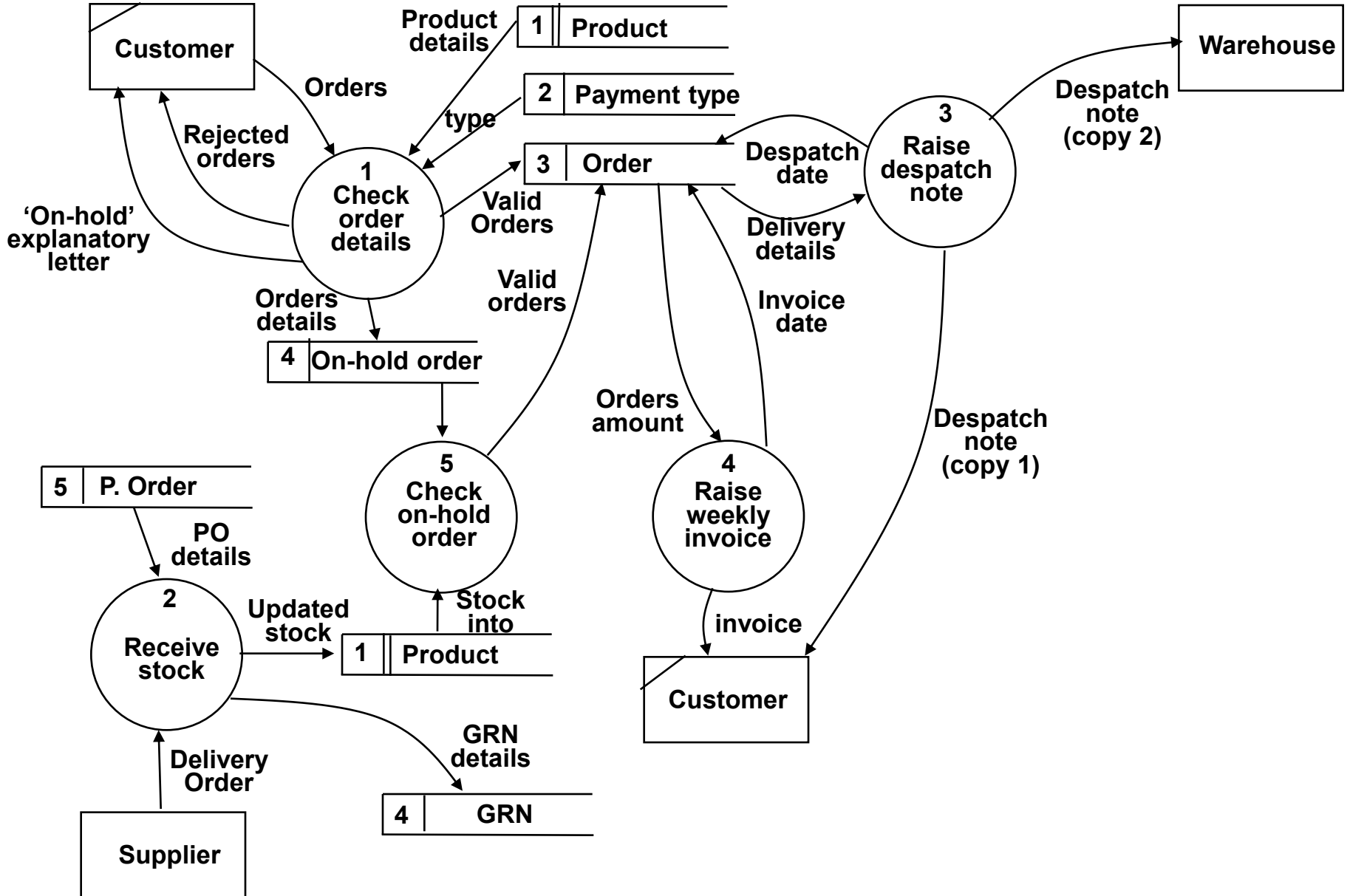
Integration Between Systems



Exercise 1 – Order Processing

- Orders received from customers are **checked** that the product and payment-type are valid values, and there is sufficient (enough) stock.
- Accepted order details are stored on an order file. If not, the rejected orders are sent back to the customer and the whole order is put on hold and an explanatory letter is sent to the customer.
- Goods are **received** from suppliers by referring to the Purchase Order (PO) and Good Received Note (GRN) is generated. Stock is updated.
- The 'on-hold' orders are **checked** and converted into an ordinary order when there is sufficient stock.
- Despatch notes (delivery note) are **raised / generated** for all orders received. A copy of the despatch note is sent to the customer and a second copy is sent to the warehouse. The despatch-date is noted on the order file.
- At the end of the week an invoice is **raised / produced** and sent to the customer. The invoice-date is noted on the order file.

Answers 1



invoice

LOGO

FROM

East Asia Trading
Pasar Pudu Baru 10
Kuala Lumpur 53000

INVOICE

MY-001

INVOICE DATE

29/01/2019

P.O.#

2330/2019

DUE DATE

24/05/2019

BILL TO

Mayang Bujang
2 Pasar Moden 55
Kuala Lumpur 55100

SHIP TO

Mayang Bujang
1721 Jln Sp 232/7
Kuala Lumpur 55103

QTY	DESCRIPTION	UNIT PRICE	AMOUNT
1	Wooden elephant figurine	600.00	600.00
2	Large cloth rice bag	45.00	90.00
3	Bamboo ladder	20.00	60.00

Subtotal 750.00

SST 6.0% 45.00

TOTAL

RM795.00



TERMS & CONDITIONS

Payment is due within 15 days

Public Bank Berhad
Account Number: 12345678
Routing Number: 0987654321098

<https://invoicehome.com/>